

Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

Sean Qin*

Department of Statistics and Data Science, Northwestern University, Evanston, IL 60208

September 2026

Abstract

Parametric Value-at-Risk and Expected Shortfall forecasts underperform in states that can be flagged in real time. A per-asset, per-day residual-diagnostic score predicts when a distribution-free tail beats GARCH-type benchmarks, an edge large in its top decile and negligible elsewhere. A jump-robust refit traces most of that edge to variance overstatement after a shock, and a smaller but significant part to residual conditional shape. The forecasts come from an amortized semiparametric estimator that transfers to newly listed assets; on the joint (VaR, ES) loss it is lowest among industry-standard benchmarks at 1% and never significantly beaten at 2.5%.

Keywords: Value-at-Risk; Expected Shortfall; misspecification; quantile regression; amortized estimation.

JEL classification: C14; C52; C53; C58; G17.

Introduction

A risk model is judged on its worst days, and those are the days on which a fixed parametric tail is most likely to be wrong. Deciding when a more flexible model is worth its added complexity is therefore a practical question.

Value-at-Risk (VaR) and Expected Shortfall (ES), the forecasts at issue, summarize the downside risk of a trading book: VaR is the lower-tail quantile of the conditional return distribution, and ES the average loss beyond it.

Three model classes produce these forecasts, and they differ in how much of the conditional return distribution they fix by assumption (Kuester et al., 2006; Nieto and Ruiz, 2016). Fully parametric filters, generalized autoregressive conditional heteroskedasticity (GARCH) and its leverage and skew extensions (Bollerslev, 1986; Glosten et al., 1993), deliver conditional scale dynamics that are hard to improve within the ARCH family (Hansen and Lunde, 2005) but read the tail off an assumed innovation density. Semiparametric filters keep that scale and estimate the innovation

*E-mail: sean.qin@u.northwestern.edu.

law from the standardized residuals, empirically (Barone-Adesi et al., 1999), through an extreme-value tail (McNeil and Frey, 2000), or by modeling the quantile directly (Engle and Manganelli, 2004), and Francq and Zakoïan (2015) supply asymptotic theory for the filtered empirical quantile. Distribution-free conditional quantile estimation by pinball-loss minimization is itself an econometric idea (Koenker and Bassett, 1978), and machine learning has scaled it up. Its members here are gradient-boosted quantile trees, neural implicit quantile networks (IQN) (Dabney et al., 2018), and the quantile-network generator of generative Bayesian computation (GBC) (Polson and Sokolov, 2023). These learners place no distributional restriction on the innovation law, but they must learn it from data and from a chosen state vector. Machine-learning models now forecast realized volatility better than heterogeneous-autoregressive benchmarks (Bucci, 2020; Christensen et al., 2023), and when pooled across a cross-section of US stocks they transfer to names outside the training set (Zhang et al., 2024). This paper asks whether a comparable advantage exists for the quantile learners studied here, whether it reaches the regulatory tail, and on which days it appears.

Industry practice sits between these classes. Under the Basel Fundamental Review of the Trading Book (FRTB), the workhorse internal models are historical simulation (HS), by far the most common method in bank disclosures (Pérignon and Smith, 2010), and its GARCH-filtered variant, filtered historical simulation (FHS) (Barone-Adesi et al., 1999). Under FRTB a bank chooses between a prescribed standardized formula and its own internal model. The buy side runs the same classical toolkit: asset managers and hedge funds, though outside Basel, forecast VaR and ES with historical simulation, parametric filters, and Monte Carlo, largely through vendor risk systems. The comparison in this paper is therefore against the methods in standard use across the industry.

Forecasts are judged by VaR exception counts at 99% and 97.5%, and the reported measure is Expected Shortfall at the 97.5% level. Reported ES enters the market-risk capital a bank holds against its book (Basel Committee on Banking Supervision, 2019), so errors are costly in both directions. A forecast that runs high ties up capital that could fund positions, while one that runs low raises the multiplier on the charge as exceptions accumulate. A method that claims to beat the industry standard must therefore beat HS and FHS on a jointly consistent (VaR, ES) score at the 97.5% level (Fissler and Ziegel, 2016), since ES alone cannot be scored (Gneiting, 2011), and it must do so under exception-test discipline (Nolde and Ziegel, 2017). Beating a vanilla GARCH on average loss is therefore neither sufficient nor trivial, because bank internal models have historically failed to beat a simple GARCH fitted to their own profit and loss (Berkowitz and O’Brien, 2002).

The advantage of the flexible model over the parametric benchmark is concentrated in particular states. On the roughly ninety percent of asset-days (one name on one trading date) when residual-shape stress is low, the estimator we use and the parametric benchmark are statistically indistinguishable on forecast accuracy, so holding the flexible model carries no detectable penalty (Section 4 reports the intervals). The gains are concentrated in the minority of days when residual-shape stress is high; averaged over all asset-days they are small, and under an annual refit schedule the average is within noise while the top-decile edge is not (Section 3.1). We ask whether an observable signal available at the forecast date can identify those asset-days in advance. It is a conditional predictive-ability question in the sense of Giacomini and White (2006): does a signal known at the

forecast date predict which of two forecasts will be more accurate on that date? We construct such a signal, a real-time misspecification score, and show that it orders the flexible model’s advantage. We then build a risk model that exploits the ordering while matching the benchmark elsewhere. Because its residual-shape model is pooled across names and conditioned on characteristics, its characteristics-only variant also delivers a forecast for an asset that has no return history at all, something no per-asset model, parametric or not, can do. Pooled estimation is known to improve volatility forecasts out of sample (Bollerslev et al., 2018; Zhang et al., 2024); here it is carried to the day-one limit and into the tail.

The score is built from the oldest specification diagnostics for a GARCH filter, the excess kurtosis and asymmetry of its standardized residuals (Bollerslev, 1987; Hansen, 1994). It is computed on a trailing window so that it is available in real time, in the spirit of the density-forecast diagnostics of Berkowitz (2001). We use these recent-residual diagnostics to predict when distribution-free quantile methods improve on parametric VaR and ES. Where residual-shape stress is high the flexible estimator has a large and significant advantage, and where it is low the advantage is statistically indistinguishable from zero in the design panel and a small positive in the point-in-time universe, with no detectable loss in either. The score orders the magnitude of the advantage and defines no sharp threshold: the edge is concentrated in the top decile but does not vanish below it.

The paper’s main contribution is the misspecification frontier: a real-time, per-asset, per-day score that predicts when distribution-free quantiles beat a GARCH- t benchmark, with significance from Diebold–Mariano tests on date-averaged loss differentials (the per-date DM of Section 3). The same kurtosis–edge gradient is traced across four universes: CRSP US equities (top-decile edge +2.98%, DM 10.5, for the pooled residual body without the EVT tail; the EVT-completed accuracy layer’s edge on the same rows is +2.48%, DM 10.1, against GARCH- t and +2.32%, DM 9.7, against GJR-GARCH-skew- t), foreign exchange, where the largest standalone wins are confined to the hyperinflation tier (pooled DM 4.12), twenty-six country indices (a developed-to-frontier gradient), and forty-three cross-asset instruments.

A jump-robust decomposition then identifies what drives that advantage. Measured against a jump-robust GARCH whose variance response to a large shock is bounded, we show that most of it is the standard filter’s post-outlier scale error, with a smaller but significant component surviving a robust scale (the top-decile +2.98% becomes +0.3–0.5%, DM 2.5–4.7). Over all asset-days the robust filter alone improves on GARCH- t by +0.42% (DM 8.47), the estimator on the standard filter by +0.35% (DM 5.35), and the estimator on the robust filter by +0.52% (DM 7.45), more than either alone (Table 2). On large caps with intraday data, re-estimation on realized-volatility residuals leaves the score’s edge within noise. On a Beta- t -EGARCH scale, whose response to a shock is bounded by construction, the flexible tail still lowers the joint (VaR, ES) loss against that model’s own Student- t tail at both regulatory levels, under frozen fits and under annual refits. This attribution, drawn on a 200-name panel and tied to a live score, is, to our knowledge, new to the flexible-versus-parametric tail-risk literature. The Korea-2026 crash, which unfolded under circuit breakers and left residual kurtosis in its ordinary range, is the negative control that sharpens the claim: a price crash is not residual misspecification, and GARCH- t wins there, as the frontier

predicts.

The pooled shape model also transfers across assets: a single fit across hundreds of names (amortized, in the machine-learning sense that one fit serves every asset) carries transferable information about conditional tail shape: it replaces per-asset estimation of the innovation shape and transfers to names never seen in training. Its gains over an own-history benchmark are largest early in a name’s life, about 9% of pinball loss at 15 to 30 trading days and 6.5–7% through the first year, measured at moderate quantiles. In its characteristics-only form it forecasts VaR and ES for assets with no return history at all, the day-one regime in which no per-asset model exists, though those first-week forecasts are not yet calibrated at the regulatory levels (Section 6).

The frontier is implemented with a semiparametric estimator that follows the two-step logic of filtered historical simulation and GARCH with an extreme-value-theory (EVT) tail (Barone-Adesi et al., 1999; McNeil and Frey, 2000). It keeps the parametric filter’s conditional scale, the component the ARCH family already estimates well once a leverage term is allowed (Hansen and Lunde, 2005), except after large shocks (Section 4), and replaces the innovation law where it fails. The forecast is therefore a GARCH conditional scale times a state-conditioned nonparametric residual quantile, with an extreme-value left tail and an optional adaptive conformal overlay. The resulting estimator and CAViaR (Engle and Manganelli, 2004) both remain in the 90% Model Confidence Set (MCS) (Hansen et al., 2011), and it improves significantly on GARCH- t , FHS, EWMA, HS, and GJR-skew- t . Its EVT-tailed accuracy layer attains the lowest joint (VaR, ES) score, the Fissler–Ziegel zero-homogeneous (FZ0) loss of Fissler and Ziegel (2016) and Patton et al. (2019), against the daily benchmarks in standard use, GARCH- t and FHS, at both FRTB levels. That ranking survives annual refitting, with exception counts near nominal through a 2008-crisis window, although the clustered exception test on that window has little power: its standard error for the accuracy layer is 0.53 points at 1%, so it does not reject any breach rate below 2.05%, and 0.28 points at 2.5%. It also beats the FZ-estimated dynamic ES models of Patton et al. (2019) and Taylor (2019), each estimated per name by FZ0 minimization, at 1% and ties Taylor at 2.5%. Two results motivate the score without supplying its mechanism. The first is the standard identity that excess pinball loss is locally quadratic in the quantile wedge, scaled by the local density (Gneiting, 2011; Ehm et al., 2016; Steinwart and Christmann, 2011), so that a wedge at a level generates regret there. The second, a tail-wedge proposition, shows how a misspecified innovation shape opens such a wedge in the tail even when the scale is right (McNeil and Frey, 2000; Francq and Zakoïan, 2015). The decomposition indicates that the score primarily captures scale error, while the surviving shape component is smaller.

The loss profile is asymmetric in the score. When the frontier is quiet the estimator ties or edges the parametric benchmark (calm-quantile edge ≈ 0). In the top decile the improvement is +2.98%, and on the hyperinflation and capital-control currencies, where the parametric assumption breaks entirely, it is 12–14%. These are reductions in a consistent scoring function for the quantile (Gneiting, 2011), not a monetary welfare measure. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level. Because the ranking of misspecified forecasts can depend on which consistent scoring function is used (Patton, 2020), the joint FZ0 score of Section 5 serves both as the regulatory comparison and as a check on that ordering under a second

score.

The paper proceeds as follows. Section 1 places the method against the industry benchmark and the forecasting literature. Section 2 develops the semiparametric estimator and the two results that motivate the frontier. Section 3 sets out the data and the forecast-evaluation protocol. Section 4 presents the misspecification frontier, and Section 5 reports the joint VaR and ES evaluation, with the ten-day extension and the cross-asset evidence in its closing subsections. Section 6 covers the use of the score and cross-sectional transfer to new listings, and Section 7 concludes.

1 Related literature and benchmark models

1.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t , the level- α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail). The return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$ for continuous F_t . In general $e_t^\alpha = \alpha^{-1} \int_0^\alpha F_t^{-1}(v) dv$, the coherent form used throughout (Artzner et al., 1999; Acerbi and Tasche, 2002). The regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 2.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at a stressed $\text{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket), while backtesting remains VaR-exception based. More than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a trading desk off the internal-models approach. The empirical bar for “beating the industry standard” is therefore joint. It requires accuracy on the joint (VaR, ES) score at 97.5% (Fissler and Ziegel, 2016; Patton et al., 2019), since ES alone cannot be scored (Gneiting, 2011), and on pinball, together with the unconditional-coverage test of Kupiec (1995) and the conditional-coverage test of Christoffersen (1998), the joint likelihood-ratio test of correct coverage and breach independence, at both levels. Significance is assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011). Real-time monitoring of VaR and ES forecast adequacy is developed by Hoga and Demetrescu (2023); the score of Section 2.5 is a cross-sectional, per-asset monitor of the same forecasts, used to rank states rather than to signal a structural break.

1.2 The parametric family, the simulation family, and CAViaR

The comparison set spans the models banks run. It includes historical simulation, exponentially weighted moving average (EWMA) variance with the RiskMetrics decay 0.94 (J.P. Morgan/Reuters, 1996), GARCH(1,1)- t (Bollerslev, 1986, 1987), and the GJR (Glosten–Jagannathan–Runkle) GARCH-skew- t (Glosten et al., 1993; Hansen, 1994), which adds a leverage term and a skewed innovation density. It also includes GARCH-FHS (Barone-Adesi et al., 1999; Hull and White, 1998), which pairs a parametric scale with empirical standardized residuals and is the filtered method that the comparison literature ranks closest to the top (Kuester et al., 2006; Nieto and Ruiz, 2016). The strongest non-nested semiparametric rival in the set is SAV-CAViaR (conditional autoregressive value-at-risk) (Engle and Manganelli, 2004), to which the comparison literature is less favorable at the 1% level (Kuester et al., 2006). The SAV (symmetric absolute

value) specification models the quantile directly by regression quantiles (Koenker and Bassett, 1978), $q_t(\tau) = b_0 + b_1 q_{t-1}(\tau) + b_2 |r_{t-1}|$ for a quantile level τ (defined in Section 2.1), the quantile analogue of an absolute-value GARCH recursion (the indirect-GARCH CAViaR is the analogue of the squared recursion). The comparison set of Table 4 does not include standalone GARCH-EVT (McNeil and Frey, 2000), which Kuester et al. (2006) rank first among classical methods. Its filter-then-tail logic is absorbed into the estimator’s own EVT stage (Section 2), so the EVT-tailed layer is its analogue within the comparison set. A same-rows comparison against standalone McNeil–Frey GARCH-EVT, per name and with a pooled tail, is reported in Sections 4 and 5 and in the Online Appendix: the conventional EVT tail carries a much smaller frontier (+0.31% per name, none with a pooled tail), and the estimator beats it by 2.64% (DM 8.9) in the top kurtosis decile and on the joint score at both levels. A single large return also distorts the GARCH filter itself, because it enters the variance recursion squared, keeps the conditional variance high for weeks, and biases the VaR and ES built on it. Carnero et al. (2007) document the effect on GARCH identification and estimation, Muler and Yohai (2008) and Boudt et al. (2013) propose robust estimators and filters, and Grané and Veiga (2014) compare corrections by their effect on risk measures. Section 4 uses a bounded-news filter of this kind to measure how much of the flexible model’s advantage is post-outlier scale error.

1.3 Distribution-free conditional quantiles and GBC

For the pooled, cross-sectional design the closest antecedent is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names. That transfer is evidence of a shared volatility mechanism, which our amortized design also exploits. For the VaR comparison the antecedent is Kuester et al. (2006), and for machine-learning volatility forecasting Bucci (2020) and Christensen et al. (2023). Deep neural-network quantile regression applied directly to VaR is the recent comparison closest to our nonparametric stage (Chronopoulos et al., 2024). Two further semiparametric VaR and ES papers in this journal are direct precedents: Jiang et al. (2022) estimate a single-index expectile model, and Luo et al. (2025) select low-frequency predictors for joint tail forecasts through a GARCH-MIDAS design with adaptive-Lasso selection. Our object differs from both. We propose no new volatility model and select no predictors; we ask when an observable signal identifies the states in which flexible tail forecasts beat parametric ones, and we decompose the source of that conditional advantage. Relative to the pooled-volatility literature, our object is state dependence in conditional tail-risk forecast performance rather than average realized-variance forecast performance, with the regulatory tail held to calibration, exception-test, and capital discipline. The question here is where a flexible model helps at all: on roughly ninety percent of asset-days it does not improve on the parametric benchmark, which cautions against extrapolating average machine-learning superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at each listing age. On tabular state, gradient-boosted trees outperform the implicit quantile network, the reverse of the intraday ranking in Zhang et al. (2024), where neural networks dominate tree-based models. At their daily horizon that dominance already disappears, and our target is a residual quantile curve rather than

realized variance, with a different feature set.

The nonparametric family estimates $Q_\phi(\tau | s_t)$ directly by pinball-loss empirical risk minimization (ERM) (Koenker and Bassett, 1978; Steinwart and Christmann, 2011). Its two members here are gradient-boosted quantile trees (GBM) (Friedman, 2001) and the implicit quantile network of Dabney et al. (2018), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklshvili et al., 2025), itself in the generalized-Bayes tradition of Jiang and Tanner (2008) and Bissiri et al. (2016). In the IQN the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time.

Trained on observed state and residual pairs, GBC’s quantile generator is a conditional quantile network, so the forward simulator of the original method is not needed and the same object applies to observational time series; Section 2.3 records how the downside-risk target changes it. The GBM is the shape model used throughout, and the IQN is retained as a benchmark. Distribution-free finite-sample coverage comes from split-conformal prediction (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019), which shifts a predicted quantile by an order statistic of held-out calibration errors so that marginal coverage holds under exchangeability with no distributional model. Adaptive conformal inference (Gibbs and Candès, 2021) replaces the fixed shift by an online update whose long-run coverage holds under arbitrary distribution shift, at the cost of the finite-sample guarantee. Barber et al. (2023) bound the coverage loss when exchangeability fails, the case for time-ordered residuals.

2 Semiparametric VaR and ES model

Each estimator below can be reimplemented from the formulas and algorithm boxes given. The Online Appendix opens with a glossary of acronyms and a schematic of the full pipeline, and states the amortized estimation loop and the monitoring procedure as Algorithms OA.1 and OA.2.

2.1 Notation and objects

Let r_t be the return of an asset on day t and $\mathcal{F}_{t-1}$ the information available before t . The conditional τ -quantile of r_t given $\mathcal{F}_{t-1}$ is $Q_t(\tau) = \inf\{q : \Pr(r_t \leq q | \mathcal{F}_{t-1}) \geq \tau\}$, the number the return falls below with probability τ . Throughout, $\tau \in (0, 1)$ is a generic quantile level and $\alpha \in \{0.01, 0.025\}$ denotes the reported regulatory levels; both index the same object. One sign convention is used throughout. The paper works in *return space*, with $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1} \int_0^\tau Q_t(v) dv \leq q_\tau$, the coherent form of Expected Shortfall (Artzner et al., 1999; Acerbi and Tasche, 2002). This is the convention of the FZ0 scoring function $(v, e < 0)$ (Fissler and Ziegel, 2016; Patton et al., 2019) and of each table, where ES values such as -6.37 are return-space conditional means. The regulatory *loss-space* numbers are their negatives, $\text{VaR}_\tau^{\text{loss}} = -q_\tau$ and $\text{ES}_\tau^{\text{loss}} = -e_\tau$, used only when quoting FRTB levels. The two are never mixed in one expression.

The *standardized residual* $z_t = (r_t - \mu_t)/\sigma_t$ is the return with the model’s conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law. The state vector s_t collects trailing realized volatility, the misspecification scores of Section 2.5, and lags, and is what the nonparametric quantiles condition on. A quantile level $\tau \sim \lambda$ is drawn from

a base distribution λ . The shape learners of Section 2.3 estimate the map from (state, level) to the residual quantile, written $q_\phi(\tau | s)$, or $H_\phi(s, \tau)$ for the network.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(x) = x(\tau - \mathbb{I}\{x < 0\}), \quad x = z - q, \quad (1)$$

whose expectation is minimized by a conditional τ -quantile (Koenker and Bassett, 1978; Gneiting, 2011)—uniquely when that quantile is unique—the way squared error elicits the mean. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below). The loss is density-free, so each estimator below dispenses with a likelihood.

Choice of scoring function. We score by pinball, a consistent scoring function for the quantile (Gneiting, 2011), at the decision-relevant levels rather than by a global proper scoring rule, for three reasons. (a) The continuous ranked probability score (CRPS), the proper scoring rule for the whole distribution (Gneiting and Raftery, 2007), is exactly the integral of the pinball loss over all levels, $\text{CRPS}(F, y) = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) d\tau$ (Gneiting and Ranjan, 2011). Scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. The quantile-weighted CRPS of Gneiting and Ranjan (2011) reweights the levels for this reason, and scoring at fixed levels is its point-mass case. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail, and CRPS would average those differences away against a body where all competitors agree. For the same reason, training uses tail-weighted τ -sampling. (b) Expected Shortfall is not elicitable on its own (Gneiting, 2011) (no loss function is minimized by the true ES alone), but the pair (VaR, ES) is, under the joint scoring functions of Fissler and Ziegel (2016), whose zero-homogeneous member, the FZ0 loss of Patton et al. (2019), is the one Section 5 uses. The FZ-scored re-evaluation of the comparison set is an extension specified in advance (Section 5). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them. Each benchmark is estimated by its own conventional criterion: Student- t or skew- t likelihood for the GARCH family, empirical residual quantiles for HS and FHS, FZ0 for the two dynamic-ES models of Section 5, and pinball for CAViaR and for the shape learners. All are then scored out of sample by the same pinball and FZ0 losses, so a model fitted on the evaluation loss carries an in-sample advantage that the out-of-sample comparison tests.

2.2 The final model: the residual-hybrid estimator

The estimator is a four-stage composition in the two-step tradition of filtered historical simulation and GARCH-EVT (Barone-Adesi et al., 1999; Hull and White, 1998; McNeil and Frey, 2000). The parametric filter estimates conditional scale, which it does well and with few parameters (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005), and the nonparametric stage estimates the state-dependent shape of the innovation law, which the fixed-shape GARCH- t benchmark fixes by assumption. The parametric route to a state-dependent shape is the autoregressive conditional density model of Hansen (1994), and Stage 2 is its distribution-free counterpart. The extreme-

value tail extrapolates beyond the estimation range using the generalized Pareto approximation of Stage 3 below. The conformal stage corrects the level of the quantile curve using its realized miss rate, updated daily, the online form of the split-conformal construction whose exchangeability-based guarantee is Proposition 1.

Motivation for the design. Conditional scale is identified by a few parameters and estimated with low variance by a parametric filter, so the shape stage is where flexible methods can add value. Per-asset tail data is scarce, which motivates pooling the standardized residuals across names. The EVT stage extrapolates the tail by the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), and the conformal stage is motivated by the split-conformal guarantee of Proposition 1, exact under exchangeability and approximate for the time-ordered residuals used here, with a coverage gap bounded by the average total-variation distance between the residual vector and its swapped versions (Barber et al., 2023). Whether the four stages together improve on the benchmark out of sample is the question of Sections 4 and 5.

Stage 1 (parametric scale). Fit GARCH- t (Bollerslev, 1986, 1987),

$$r_t = \mu + \sigma_t z_t, \quad \sigma_t^2 = \omega_G + \alpha_G(r_{t-1} - \mu)^2 + \beta_G \sigma_{t-1}^2, \quad z_t \sim t_\nu \text{ (standardized)}, \quad (2)$$

by Student- t quasi-maximum likelihood on the estimation window of Section 3 (annually refit in ongoing use), and keep $\hat{\mu}$, $\hat{\sigma}_t$, and the residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$. The t_ν likelihood is used to estimate conditional scale; Stages 2 and 3 re-estimate the residual shape. This two-step design, a parametric filter followed by a residual quantile, is the one whose asymptotics Francq and Zakoian (2015) derive for an empirical residual quantile, and McNeil and Frey (2000) fit the same filter by Gaussian pseudo-likelihood while explicitly declining to believe the innovation law it assumes. No comparable result covers a pooled tree learner in the second stage, so the paper’s inference is on the forecasts (Section 3.3).

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\hat{\phi} = \arg \min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau \sim \lambda} \rho_{\tau}(\hat{z}_{j,t} - q_{\phi}(\tau \mid s_{j,t})). \quad (3)$$

Stage 3 (EVT tail). In the sparsely observed lower tail, where (3) is fitted on few exceedances, use the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Balkema and de Haan, 1974; Pickands, 1975; McNeil and Frey, 2000). The finite tail mean of the generalized Pareto distribution (GPD) requires $\xi < 1$, a condition each fitted tail in this paper satisfies by a wide margin.¹ On losses $X = -\hat{z}$, set the threshold u at the empirical $(1 - p_0)$ quantile of the pooled training residual losses, with $p_0 = 0.025$. Fit the GPD $G_{\xi, \beta}(y) = 1 - (1 + \xi y/\beta)^{-1/\xi}$ to the excesses $y = X - u > 0$ by maximum likelihood with location fixed at zero, strictly on past data, giving $(\hat{\xi}, \hat{\beta})$; (4) is evaluated at these estimates and hats are suppressed below. The shape ξ sets how heavy the tail is and the scale $\beta > 0$ how wide; this β is the GPD scale, distinct from the GARCH

¹The peaks-over-threshold formulation uses tail data more efficiently than block maxima (Embrechts et al., 1997) and is standard in risk management. Here $\xi > 0$ is the heavy-tailed case.

persistence β_G . One (ξ, β) pair is fitted per panel and held fixed across states and levels within a fit. For $\tau \leq p_0$, set

$$\hat{q}^{\text{EVT}}(\tau) = -\left[u + \frac{\beta}{\xi}((\tau/p_0)^{-\xi} - 1)\right], \quad (4)$$

which is continuous at its own endpoint, $\hat{q}^{\text{EVT}}(p_0) = -u$. The formula requires $1 + \xi y/\beta > 0$, and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \rightarrow 0$ exponential limit. The threshold u is unconditional with respect to the state, while the Stage-2 body quantile at p_0 is the state-conditional $\hat{q}_\phi(p_0 | s)$, so the two branches need not meet at the join. In the tail the forecast takes the pointwise minimum of the two, the more conservative (more negative) branch, so the pooled GPD extrapolation can tighten a thin state-conditioned body on a stressed day but never loosen it. The curve is then made non-crossing by monotone rearrangement, which cannot move it further from the true quantile curve in any L^p norm (Chernozhukov et al., 2010). Both steps are written out in (5).

Stage 4 (conformal finish). The finished curve is shifted per level by a scalar, $\tilde{Q}_t^z(\tau) = Q_t^z(\tau) + c_\tau$ with Q_t^z the envelope of (5), and the shift is the last operation, after the minimum and the rearrangement. In its static form c_τ is fixed once: on a held-out calibration split of size n , compute the signed errors $\varepsilon_i = \hat{z}_i - Q_i^z(\tau)$ of the envelope and let c_τ be their $\lceil (n+1)\tau \rceil$ -th order statistic (Vovk et al., 2005; Lei et al., 2018), the one-sided form of conformalized quantile regression (Romano et al., 2019). In the adaptive form used here the shift starts at zero and is updated each day from the panel breach frequency $b_{\tau,t}$ of the shifted forecast, $c_{\tau,t+1} = c_{\tau,t} - \gamma(b_{\tau,t} - \tau)$ (Gibbs and Candès, 2021), so the shift in force on day $t+1$ uses only breaches observed through day t . The *accuracy layer* that carries the FZ0 comparisons of Section 5 and Table 4 sets $c_\tau \equiv 0$; the *adaptive overlay* applies the update at both regulatory levels with $\gamma \in \{0.02, 0.05\}$, and in era it moves the 97.5% breach rate toward nominal at no cost in FZ0 (Section 5). The coverage statement of the static form is standard (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019) and is restated here for a quantile model $\hat{q}_\phi$ and its calibration errors.

Proposition 1 (split-conformal validity under exchangeability (standard)). *If the calibration and test errors $\varepsilon_1, \dots, \varepsilon_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \leq n$ the shifted quantile satisfies*

$$\tau \leq \Pr(z_{n+1} \leq \hat{q}_\phi(\tau | s_{n+1}) + c_\tau) = \frac{\lceil (n+1)\tau \rceil}{n+1} < \tau + \frac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of ε_{n+1} among $\{\varepsilon_1, \dots, \varepsilon_{n+1}\}$ is uniform on $\{1, \dots, n+1\}$. The event $z_{n+1} \leq \hat{q}_\phi(\tau | s_{n+1}) + c_\tau$ equals $\{\varepsilon_{n+1} \leq \varepsilon_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \leq \lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil / (n+1) \in [\tau, \tau + \frac{1}{n+1}]$ (Vovk et al., 2005). The full argument is in the Online Appendix, together with the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1) and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$. The guarantee is marginal over the state s_{n+1} rather than state-conditional. $\square$

The proposition's scope is narrower than its use. In practice the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable, so the proposition only motivates the construction. The static shift fails here for that reason, since the calibration block and the 2020–2024 test era miss on opposite sides of nominal and a shift fitted on one over-covers the

other (Section 5). Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018; Barber et al., 2023; Gibbs and Candès, 2021), and the adaptive update above is that of Gibbs and Candès (2021), which gives up the finite-sample statement for long-run coverage under arbitrary distribution shift. Whether it is switched on was decided by the exception tests and the joint score under a rule fixed before the run (Section 5).

One further implementation gap: the first-stage GARCH is fit on the full pre-test history, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer comparisons never read the calibration split and share their filter with the benchmark, so their information sets match. The adaptive overlay reads only breaches already observed in the test period, so no reported forecast reads the calibration split. Re-estimating with the filter stopped at the calibration boundary leaves the accuracy advantage intact (DM 5.0 and 5.1). The strict estimator’s gap to the full-window benchmark (-0.7 , -2.6) matches that benchmark’s own margin over its short-window twin (2.8 , 3.8), so the gap is the cost of a quarter less estimation data and shows no leakage.

The full forecast composes the stages:

$$Q_t^z(\tau) = R\left[\min\{\hat{q}_\phi(\tau | s_t), \hat{q}^{\text{EVT}}(\tau)\} \mathbb{I}\{\tau \leq p_0\} + \hat{q}_\phi(\tau | s_t) \mathbb{I}\{\tau > p_0\}\right], \quad \hat{Q}_t(\tau) = \hat{\mu} + \hat{\sigma}_t(Q_t^z(\tau) + c_\tau), \quad (5)$$

where $R[\cdot]$ is the monotone rearrangement across levels (Chernozhukov et al., 2010) and $c_\tau \equiv 0$ in the accuracy layer. Q_t^z is the standardized-residual quantile curve, the single object from which both risk measures are read; the conditional scale $\hat{\sigma}_t$ carries the day’s volatility from the GARCH stage. The return-space forecasts are $\hat{q}_{\tau,t} = \hat{Q}_t(\tau)$ and $\hat{e}_{\tau,t} = \hat{\mu} + \hat{\sigma}_t(\tau^{-1} \int_0^\tau Q_t^z(v) dv + c_\tau)$, the latter a numerical integral of the same curve. The overlay applies one scalar per level to the VaR node and to the ES integral at that level, so the shift sits outside the integral by construction; the accuracy layer sets $c_\tau \equiv 0$ and the two coincide. The body branch is the minimum on about 40% of tail nodes. ES_α is the integral of (5) over $(0, \alpha]$, computed to convergence: the body is interpolated across sixty fitted levels on $[\alpha/40, \alpha]$, the GPD branch is evaluated exactly at each of 2,000 nodes, and below $\alpha/40$, where the GPD quantile diverges, its integral is taken in closed form. A 20-node midpoint rule, used in earlier versions, understates the engine’s $|\text{ES}|$ by about 1.6% for every one of the 200 names and distorts the ES backtests (Online Appendix). With the GPD closed form of McNeil and Frey (2000) in place of the integral, the accuracy layer’s FZ0 is significantly worse at both levels (DM 2.5 at 1% and 4.2 at 2.5%) and its margin over GARCH- t at 2.5% narrows from DM 5.5 to 4.0 (Online Appendix). The splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$): DM 4.5–4.7 at the 1% level and 4.2–5.5 at 2.5%.

Industry models as special cases. Equation (5) nests the standard toolkit, in the taxonomy of Kuester et al. (2006). Historical simulation sets $\hat{\mu} \equiv 0$, $\hat{\sigma}_t \equiv 1$, and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS) (Barone-Adesi et al., 1999; Hull and White, 1998) takes q as the unconditional empirical quantile of past *residuals*, i.e. Stage 2 with the state dependence deleted. GARCH- t (Bollerslev, 1987) takes $q = t_\nu^{-1}$, a fixed parametric shape. Whatever FHS captures, (3) can represent, and it adds conditional shape. CAViaR (Engle and Manganelli, 2004) sits outside the nesting. It autoregresses on the quantile directly, with no scale/shape split, and is the benchmark with which the estimator ties (Section 5).

Algorithm 1 (residual-hybrid estimator). (i) Fit (2) per asset on the estimation window. Store $\hat{\sigma}_t, \hat{z}_t$. (ii) Pool $(s_{j,t}, \hat{z}_{j,t})$ across names. Train the shape model by (3). (iii) Fit the GPD tail (4) by maximum likelihood on the pooled training excesses. Take the minimum with the body at $\tau \leq p_0$ and rearrange. (iv) Update the shift $c_{\tau,t+1} = c_{\tau,t} - \gamma(b_{\tau,t} - \tau)$ daily from the panel breach frequency, starting at zero (overlay only; $c_\tau \equiv 0$ in the accuracy layer). (v) Forecast by (5). In *ongoing use* all stages refit on an annual walk-forward schedule. The paper’s *evaluated* protocol freezes each stage once per split, the more conservative test since no stage ever adapts to the test era. Section 3.1 verifies the frontier separately under the true annual-refit schedule.

2.3 Estimating the shape: gradient boosting and neural quantile generation

Two estimators realize $q_\phi(\tau | s)$ in (3). The trees carry every reported forecast; the network, the architecture at the core of GBC, is reported as a benchmark.

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta h_m(\cdot)$ is grown stagewise by gradient boosting (Friedman, 2001), where $\eta = 0.06$ is a fixed shrinkage rate, $M = 250$ trees of depth three are grown, and h_m is the m -th regression tree, fitted by least squares to the pinball pseudo-residuals. Each new tree h_m is fit to the negative gradient of the pinball loss, which equals τ_k where the current model under-predicts and $\tau_k - 1$ where it over-predicts. The fitted grid is made non-crossing by monotone rearrangement, which sorts the values $\{q(\tau_1|s), \dots, q(\tau_K|s)\}$ in τ and is weakly closer to the true monotone quantile curve in every L^p norm (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest point estimator, about 0.75% better pinball than the tuned network, consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN), the network at the core of GBC. A τ -conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s, \tau) = g(\psi(s) \circ c(\tau)), \quad c_j(\tau) = \text{ReLU}\left(\sum_{i=1}^{n_c} \cos(\pi i \tau) w_{ij} + b_j\right), \quad (6)$$

where $\circ$ is elementwise multiplication, $c(\tau)$ is a cosine embedding of the level with $n_c = 64$ features, and g, ψ are two-layer ReLU feed-forward nets of width 192 (128 in the untuned run) acting on the level embedding and on the standardized state respectively. The tuned variant uses tail-aware τ -sampling, monotone rearrangement of the τ -curve, and a conformal shift. The tail-aware sampling, $\lambda = \frac{1}{2}U(0, 1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$, was set once, before the tuned run, so that about a tenth of the training levels fall below 2.5% against 2.5% under uniform sampling; the mixture and the Beta parameters were not tuned on any test data. It answers the untuned run, whose 99% breach rate was 3.2% under uniform sampling, and the tuned IQN row of Table 5 is a design revision made after that run and evaluated on a later sub-window. The trade-off between the two estimators is set out under *Estimator choice* below.

How training works. The network minimizes the empirical version of (3) by stochastic gradient descent, each example drawn at a fresh level $\tau_{jt} \sim \lambda$ each epoch, so one set of weights carries the entire quantile curve. Training runs for a fixed budget of 12,000 minibatch steps (Adam, batch 512, rate 10^{-3} ; 6,000 in the untuned run) with no early stopping. Monotone rearrangement and the

conformal shift are applied *after* training, and the whole pipeline is refit on the annual walk-forward schedule using only past data. Layer sizes, learning rates, and seeds are reported in the replication package.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names, and it works, at cold-start in particular, because the mapping is estimated across names instead of within one. A per-asset GARCH learns *name-specific parameters* from that name’s own history, so it has no forecast until a history exists. The amortized model instead learns a single mapping from observable state to residual shape, estimated from millions of asset-days across names. For a newly listed asset, the estimated mapping can be evaluated from the available state variables without per-asset refitting. As the name’s own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically.

Feature ablation at full scale (1.32M rows, 720 names, 288 held out), with the characteristics built point-in-time from data through the previous day (trailing 250-day volatility and market beta; market capitalization, which the returns panel cannot rebuild, is dropped), shows the transfer edge spread across the name’s own recent dynamics and its characteristics. Removing short-window realized volatility costs +0.54% and removing all cross-sectional characteristics +0.45%, while the characteristics-only form is 2.2% worse than the full model. Characteristics matter at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an affine own-history shrinkage of the amortized quantile toward the name’s trailing location and scale, with weight $\text{age}/(\text{age}+80)$, both hurt at each listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly.

On held-out names the amortized model has lower cross-name mean pinball loss than the own-history benchmark on 91.7% of the 2,767 test dates, and lower mean loss for 99.0% of the names. The benchmark is each name’s expanding-window mean and standard deviation with a Gaussian quantile, scored at seven levels from 0.05 to 0.95, so the transfer result concerns distributional shape at moderate quantiles; that loss contains no 1% or 2.5% level.

Estimator choice. The two estimators minimize the same loss over different index sets. The trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\hat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level, while the network solves $\min_{\phi} \sum_i \mathbb{E}_{\tau \sim \lambda} \rho_{\tau}(\hat{z}_i - H_{\phi}(s_i, \tau))$, one model for the continuum. At the fixed regulatory levels the trees are preferable: best point accuracy and no GPU, with the grid made monotone by rearrangement. The network learns one continuous conditional quantile map at a measured cost of $\sim 0.75\%$ in point accuracy. The two share each upstream and downstream stage of (5), so swapping one for the other changes a single component of the pipeline.

Relation to generative Bayesian computation. Relative to the generative Bayesian computation of Polson and Sokolov (2023), the downside-risk target changes the estimator in five ways, each introduced above. The sampling measure λ is tail-weighted rather than uniform, because uniform sampling starves the extreme levels that downside risk depends on. The object modeled is the GARCH-standardized residual rather than the raw return, so that the conditional-scale dynamics the parametric model already estimates well are kept. Beyond the training support, extrapolation is handed to the GPD tail (4) rather than to the learner. The conformal stage adds a coverage correction driven by realized breaches. And the amortization runs across the real cross-section

of hundreds of assets instead of across simulations of a single model, which is what produces the cold-start forecasts and transfer results. When such machinery improves on the parametric filter is not a question generative computation addresses, and the score of Section 2.5 is built to answer it.

2.4 Formal results: why shape misspecification must show up in the loss

The nonparametric edge concentrates where the trailing residual-shape diagnostics are extreme, an empirical finding with a structural motivation in the pinball loss. Two elementary results, with proofs in the Online Appendix, supply it; they motivate looking where the shape diagnostics are extreme but do not prove that mk_{63} estimates the wedge, a link established empirically.

Lemma 1 (pinball regret identity (standard)). *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q , finite by the moment condition. Then for any q ,*

$$S(q) - S(q_F) = \int_{q_F}^q (F(v) - \tau) dv \geq 0, \quad (7)$$

with equality iff $F(u) = \tau$ for Lebesgue-almost each u between q_F and q . If F has density bounded as $0 < m \leq f \leq M$ on that closed interval,

$$\frac{m}{2} (q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2} (q - q_F)^2. \quad (8)$$

Proof. In the Online Appendix. □

The extra loss a misspecified forecaster pays is thus locally quadratic in its quantile error, and the frontier claim reduces to whether shape misspecification opens a quantile wedge at the levels that matter. For a fixed Student- t shape that is too thin (McNeil and Frey, 2000; Kuuster et al., 2006), it does:

Proposition 2 (tail wedge under tail-shape misspecification (elementary)). *Let $Q_\nu(\tau)$ denote the τ -quantile of the unit-variance Student- t with $\nu > 2$ degrees of freedom. For each pair $\nu_2 > \nu_1 > 2$ there exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2]$ such that $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$ for all $\tau < \tau^*$. The fatter-tailed law thus has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \rightarrow \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2} (Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ at each level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof. In the Online Appendix. □

The tail-wedge motivates but does not establish the empirical score-ordering result. We claim no monotonicity theorem at the regulatory levels: because the innovations are unit-variance, a heavier tail can carry the *less* extreme 2.5% quantile, with a pair-dependent crossover τ^* (0.018 for t_3 against the normal, 0.023 for t_5 against t_8), so 1% sits below it and 2.5% just above (Patton, 2020). The bound is a fixed-level statement rather than a tail-limit one: because $m = m(\tau)$ shrinks as

$\tau \downarrow 0$, the certified regret $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ stays positive at each level but does not grow into the tail, and in fact vanishes there. We therefore treat the score-ordering as an empirical regularity, motivated by the tail-wedge mechanism rather than implied by it and documented across the design panel, the frozen holdout, the FRTB comparison, and the cross-market universes.

2.5 The misspecification score

We use “misspecification score” as shorthand for a diagnostic of local residual-shape stress under the fitted GARCH- t benchmark. It is a trailing-window moment monitor in the spirit of the PIT-based density-forecast diagnostics of Diebold et al. (1998) and Berkowitz (2001), used as a rank rather than as a formal specification test.

GARCH- t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student- t with fixed shape (Bollerslev, 1987). Static heavy tails do not violate this, since ν absorbs them. A locally wrong shape does: recent residuals far fatter-tailed or more asymmetric than any fixed t , the time variation in conditional skewness and kurtosis that Hansen (1994) models parametrically. The score is therefore computed per asset-day over the trailing window $W_t = \{t - 63, \dots, t - 1\}$ of $n_w = 63$ residuals, with window mean $\bar{z}$ and standard deviation s_z . The window *ends the day before the forecast target*, so each signal is lagged at least one day relative to the loss it is paired with. The frontier sorts of Section 4 pair S_{t-1} with the day- t loss, an alignment enforced by construction. The two moment statistics are

$$\text{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\hat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \quad \text{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\hat{z}_i - \bar{z}}{s_z} \right)^3, \quad (9)$$

In estimation, mk_{63} and sk_{63} are computed as the bias-corrected sample excess kurtosis and skewness. At the full window length the bias correction is a common affine rescaling of (9), so the pooled decile ranks that the frontier uses are unchanged. *Skewness* is the signed third moment sk_{63} . *Asymmetry* always means its absolute value $|\text{sk}_{63}|$, departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-5, \dots, t-1\}} |\hat{z}_i|$. The frontier orders the edge by the decile of each of these signals. The composite score is the maximum of their trailing percentiles, $\text{score}_t = \max\{\hat{F}_t(\text{mk}_{63}), \hat{F}_t(|\text{sk}_{63}|), \hat{F}_t(J_5)\}$, where $\hat{F}_t$ is each signal’s percentile in the pooled test panel for the sorts of Table 1 and Figure 1, and in the trailing panel available before t for the expanding-cutoff check reported beside them. The composite is itself deciled. All three are predetermined (trailing-window) and computable on any asset with a fitted GARCH, so the frontier can be monitored in real time. Online Appendix Table OA.3 reports the associated per-decile actions.

Why these statistics. Conditional on the fitted location-scale specification (μ_t, σ_t) , the distributional restriction the flexible family relaxes is the fixed t_ν innovation *shape*. The lowest-order ways a fixed symmetric law can be locally wrong are the third and fourth standardized moments (Bollerslev, 1987; Hansen, 1994). The statistic sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies. An exact t_ν sample has known positive excess kurtosis, but the score is used as a rank, so only its cross-sectional and temporal variation matters, not its t_ν baseline. J_5 catches jump episodes too recent for the window moments

to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent, and Section 4 shows that the realized edge concentrates there.

3 Data and evaluation protocol

Equity panels come from CRSP (the Center for Research in Security Prices) daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations. Each modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: the top 200 common shares (share codes 10 and 11) by average market capitalization among those with at least 3,000 observations in the window. Returns are in percent and include CRSP delisting returns where present. A name whose delisting return CRSP does not record ends at its last regular return, an omission Shumway (1997) shows to be large and negative for performance-related delistings. The one delisting event in the design panel moves the main results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data.

The replication repository carries code, configuration, and derived statistics, and the panels can be rebuilt from the queries above by any licensed subscriber. The five evaluation panels are a 2014–2024 design equity panel (200 CRSP names, 221,600 test observations) for the frontier and estimator, a frozen-spec 2000–2013 holdout (200 names, 280,608 observations), a 140-name subpanel for the neural-network comparison and auxiliary checks, and Bloomberg cross-asset (43) and country-index (26) sweeps. The Online Appendix tabulates them.

Splits are chronological within each name. The first 60% of a name’s history is the estimation window and the last 40% is test. The final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH- t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on the pooled training-window residuals (the estimation window less the calibration split), the GPD tail on the same pooled residuals below their empirical 2.5% point. The adaptive shift of Stage 4 starts at zero and is updated from test-period breaches as they are observed. No other stage is updated inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule’s survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the note that closes this section. Distributional accuracy is scored by mean pinball loss across a fixed grid of quantile levels, twelve in the FRTB comparison of Table 4 and eleven in the frontier sorts, a consistent scoring function for the quantile vector that is the point-mass case of the quantile-weighted CRPS (Gneiting and Ranjan, 2011), and joint (VaR, ES) pairs by the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019). Loss comparisons use per-date Diebold–Mariano statistics (Diebold and Mariano, 1995) with a Newey–West variance (Newey and West, 1987), and multi-model fields use the Model Confidence Set (Hansen et al., 2011). Exception behavior is tested by the Kupiec unconditional-coverage,

Christoffersen conditional-coverage, and date-clustered breach tests at the two regulatory levels.

“Frozen-specification” has a specific meaning here: a run is frozen-specification when the full specification (signals, features, hyperparameters, split rule, and test statistics) is fixed and the predictions are written down before the data are pulled. We do not use the word *registered*, since no third-party registry or timestamp exists for them. The holdout pipeline is line-for-line the design-era pipeline, and both are public in the replication repository, so the two can be compared directly to confirm that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 4 ran under this rule. As a post-hoc check, the composite score also replicates on this holdout. Its top decile carries +1.12% (DM 2.51), marginally above its kurtosis component (+1.04%, DM 2.16, on the residual-hybrid learner; the frozen returns-space learner of Section 4 gives +0.89% at the same DM), and +0.71% (DM 2.08) under the expanding prior-only cutoff. Because the composite was not the rule frozen at the time, this is replication on an already-examined sample, not a frozen-specification test.

3.1 Common shocks, calendar overlap, and the pooled learner

Two nested flexible specifications appear in the paper. The residual-hybrid holds the GARCH conditional scale fixed and models the standardized residuals; it produces the design-panel frontier (Section 4) and every VaR and ES forecast. A simpler returns-space variant, plain enough to fix in advance, carries the pre-committed out-of-sample and robustness checks. Because a 60/40 per-name split can place one name’s training window on another’s test dates (López de Prado, 2018), a pooled learner could meet a systemic event in training that a per-name model meets only out of sample. A strict calendar-time split closes that channel (Gu et al., 2020): each stage is re-estimated strictly before 2020-01-01 and tested after, and the top-decile edge survives (+2.45%, DM 6.03). It also survives a pre-2007 cut that removes the 2008 crisis from every training window (+0.29% overall, DM 3.08), and an annual walk-forward that refits the GARCH, residuals, score, and learner at each January-1 cutoff, under which the top decile still carries +2.12% at DM 4.41 while the average edge washes out (−0.22%, DM −0.59); the returns-space learner of the frozen holdout specification gives +2.47% (DM 6.05) and +0.32% (DM 0.8) under the same schedule. Stale benchmark parameters therefore do not produce the conditional result. The overlap mechanism and the full calendar-split, pre-2007, and walk-forward tables are in the Online Appendix.

3.2 Cross-sectional dependence, family-wise error, and the universe rule

Each main test collapses the cross-section to one number per date before any variance is estimated, and the bootstraps resample whole dates instead of asset-days, following the pooled-average panel test of Qu et al. (2024). Over the full 3×10 signal-decile grid, Romano–Wolf step-down family-wise control (Romano and Wolf, 2005, 2016) leaves nine cells significant at 5%, all of them frontier cells and including the top deciles of all three signals; any cell that does not survive is reported as indistinguishable from zero wherever it is discussed. The exploratory single-instrument universes carry their own Holm adjustment over 93 comparisons (Holm, 1979). Survivorship is addressed by a point-in-time re-selection that fixes the universe at the top 200 names as of Q1-2000, applies no minimum-history filter, and rides each delisting return to the end: on the 164 names that clear the

fitting floors the overall edge is +0.61% (DM 9.16) and the frozen top bucket carries +3.17% at DM 7.0, larger than in the survivor-tilted holdout. The date-bootstrap construction, the cell-by-cell adjustment, and the point-in-time details are in the Online Appendix.

3.3 Reporting and inference conventions

The reporting and inference conventions used in the empirical sections that follow are fixed here. Edges are percentage reductions in pinball loss, so larger is better for the named model. A “+2.7% edge” means the model’s average loss is 2.7% lower than the benchmark’s. Ratios are model-over-benchmark loss ratios, so below 1 favors the model and $1 - \text{ratio}$ is the edge ($0.973 = \text{a } 2.7\% \text{ win}$, $1.043 = \text{a } 4.3\% \text{ loss}$).

DM is the Diebold–Mariano t -statistic on the loss difference. The forecasts are frozen objects, so the comparison is of forecasts rather than of estimated models, the use Diebold (2015) endorses. Estimation is part of each forecasting method, and the conditional predictive-ability framework of Giacomini and White (2006) holds for any estimator, parametric or not, under the fixed estimation scheme of every headline comparison; the annual-refit runs use expanding windows, which that framework does not cover, and are reported as robustness checks. For the per-date DM, losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$. The orientation is stated at each use (benchmark minus the residual-hybrid in its main comparisons, so that positive favors the residual-hybrid). The test is then run on the resulting $\sim 1,100$ -date series with a Newey–West (lag-10) variance (Newey and West, 1987), so the effective sample is the number of dates, not the number of asset-days. The lag replaces the $h - 1 = 0$ truncation of Diebold and Mariano (1995) because misspecified one-step forecasts leave serially correlated loss differentials.

Reported p -values are one-sided in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%), a direction fixed before the run for the residual-hybrid’s main comparisons, and where a winner is instead chosen by the data the one-sided level reads as a two-sided level of twice its size. A t -statistic above 2.6 is significant under either convention, and marginal cases are flagged where they occur. A tie means the per-date DM cannot reject zero at the two-sided 10% level. The point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible we make it as one. For the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin, which is a two-one-sided-tests conclusion rather than a failure to reject.

The Model Confidence Set is computed on the same per-date loss matrix with the range statistic of Hansen et al. (2011) and a stationary bootstrap over dates (Politis and Romano, 1994) (mean block length 10 days, $B = 1,000$ replications in the main comparison, $B = 800$ in the CAViaR extension). Hansen et al. (2011) implement the set with a fixed-length block bootstrap (Künsch, 1989), and the stationary variant is the one used in the SPA test of Hansen (2005). Breach rates are realized exception frequencies against the stated nominal level, so a rate above nominal is under-coverage of risk and a rate below it is conservative. Correlations quoted for the frontier relate a universe’s residual-kurtosis level to the size of the nonparametric edge across that universe’s members. No other significance marks are used.

Table 1: The misspecification frontier: nonparametric edge over GARCH- t by decile of the trailing misspecification score (200 CRSP names, 1,108 dates, 221.6k asset-days). Edge = (GARCH pinball – nonparam pinball)/GARCH pinball.

Decile	Edge by signal decile (%)		
	mk ₆₃ (kurtosis)	sk ₆₃ (asymmetry)	jump ₅
1–8 (bulk)	≈ 0	≈ 0	mixed
9	+0.46	+0.59	—
10 (top)	+2.98	+2.78	+1.96
Overall	+0.35		

Per-date Diebold–Mariano: top mk₆₃ decile +2.98%, DM 10.5; |sk₆₃| +2.78%, DM 9.85; jump₅ +1.96%, DM 7.63 (all $p \approx 0$). Overall +0.35%, DM 5.35; the composite’s bottom decile is indistinguishable from zero (–0.00%, DM 0.36). The composite score’s top decile carries +2.79%, DM 8.91 (Figure 1). The nonparametric forecast in this table is the pooled residual body of Stage 2 without the EVT tail; the EVT-completed accuracy layer of Section 5 carries +2.48% (DM 10.1) in the top kurtosis decile on the same rows (Online Appendix Table OA.10). Top-decile signal means: kurtosis ≈ 25 , |skew| ≈ 4 . jump₅ is U-shaped, its lowest decile also carrying an edge (+1.17%, DM 5.5), so its bulk is mixed rather than near zero.

4 The misspecification frontier

4.1 The within-universe result

On 200 CRSP names across 1,108 test dates (≈ 221.6 k asset-days), we bucket the per-day pinball edge of the residual-hybrid over GARCH- t into deciles of each misspecification signal. Both forecasters share the GARCH conditional scale, so any advantage enters through the residual quantile. Because that quantile is conditioned on trailing realized volatility as well as on the score, it can also act as a state-dependent correction to the daily scale, and Section 5 shows that on large capitalizations a realized-variance scale absorbs much of this component (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005). The frontier therefore measures the value of a state-conditioned residual law given a daily parametric scale, not a pure innovation-shape effect. Sorting forecast performance by a predetermined state is a conditional predictive-ability comparison in the sense of Giacomini and White (2006). Table 1 reports the sort on which the rest of the paper builds.

The edge is concentrated in the top decile: the top mk₆₃ decile carries +2.98% at per-date DM 10.5, the overall edge is +0.35% at DM 5.35, and the lower deciles are statistically indistinguishable from zero. The sort has a regression form. Taking the asset-day loss differential (GARCH- t pinball minus engine pinball) and the lagged score as the conditioning variable, the conditional-predictive-ability test of Giacomini and White (2006) with instruments (1, score) rejects equal conditional accuracy (Wald $\chi^2_2 = 73.6$ on the composite percentile, 90.0 on the kurtosis percentile), and the slope of the differential on the score is positive with a date-clustered t of 8.6 (composite) and 9.8 (kurtosis). On a date-by-date reading the engine has the lower loss on 74% of dates in the top decile and 55% in deciles one through nine.

GARCH-EVT on the same rows. The frontier is a statement about a state-conditioned residual shape, so we ask whether a conventional unconditional tail captures it. On the rows of Table 1, a McNeil–Frey GARCH-EVT forecast (the same GARCH- t filter, GPD tails fitted by maximum likelihood to the 10% most extreme training residuals per tail, empirical residual quantiles in between) shows no frontier of the estimator’s size. Against GARCH- t its top-decile edge is +0.31% (DM 3.6) per name and −0.16% (DM −2.7) with a pooled tail, and its overall edge is negative in both forms. The estimator beats the pooled GARCH-EVT by +2.64% (DM 8.9) in the top kurtosis decile and by +0.10% (DM 2.5) in deciles one through nine, and the per-name GARCH-EVT by +2.18% (DM 9.6) and +0.21% (DM 4.6). GJR-GARCH-skew- t , which adds a leverage term to the filter, shows no frontier either (top-decile edge over GARCH- t +0.16%, DM 0.6), and the accuracy layer beats it by +2.32% (DM 9.7) in the top decile. The same-rows table and the GPD threshold diagnostics are in the Online Appendix, together with the rest of the comparison set on the same rows (Table OA.10). Of the standard benchmarks, only SAV-CAViaR shows a top-decile edge over GARCH- t above one percent (+2.81%, DM 8.2). It models the quantile directly and reacts to $|r_{t-1}|$, so it carries the same frontier the engine does, and the two are within a third of a percent in every region (top decile −0.33%, DM −1.8, marginal under the convention of Section 3.3; overall −0.12%, DM −1.6, engine over CAViaR), the tie Section 5 reports on the FRTB battery. The conditional-predictive-ability slope of each benchmark’s loss differential against the engine on the lagged score is positive with $t > 2$ for GARCH- t , per-name GARCH-EVT, pooled-tail GARCH-EVT, pooled FHS, per-name FHS, rolling FHS, EWMA and GJR-GARCH-skew- t , so the engine’s advantage over each of them grows with the score; it is flat for historical simulation and SAV-CAViaR.

The composite score. Table 1 sorts on each signal separately. The monitor the paper uses is the composite score, the maximum across the three signals’ trailing-panel percentiles (kurtosis, asymmetry, jump). Evaluated directly, its top decile is a 10% cell carrying +2.79% (DM 8.9), and the lower nine deciles sit between −0.3% and +0.35%. The edge survives the strictly non-anticipating expanding cutoff at +2.72% (DM 9.0). The three components each carry a top-decile edge on their own: +2.98%, +2.78%, and +1.96% (kurtosis, asymmetry, jumps). The maximum of decile ranks instead flags the union of the three top deciles, a 19% cell at +1.72%, so we report the re-deciled composite.

The score’s content could reflect model-relative misspecification, departure from each name’s own fitted t_{ν_i} , or the absolute level of recent tail activity. Converting each mk_{63} into a percentile of the simulated null of 63-window kurtosis under the name’s fitted t_{ν_i} leaves the frontier essentially unchanged (top decile +3.0%, DM 8.5, against +2.98%, DM 10.5 raw). The normalization is nearly rank-preserving, however (Spearman 0.97, 82% top-decile overlap), so it cannot by itself separate the two.

A discriminating test instead favors the raw-tail-activity reading. In a within-date cross-sectional (Fama–MacBeth) regression of the per-observation edge on the raw score and the ν -relative percentile, the raw score carries the predictive content ($t = 7.8$), while the ν -relative percentile has significantly negative incremental content ($t = -5.2$). Double-sorting agrees. The top-raw-kurtosis-quintile edge falls from +1.9% for the most fat-tailed (low- ν) names to +1.5% for

the thinnest-tailed, and observations high in the ν -relative score but not in raw kurtosis carry no edge (+0.3%, DM 0.5). The signal is thus the absolute level of recent tail activity rather than a per-name ν -normalized surprise. The signal still measures shape stress relative to the fitted model, since mk_{63} is computed on the standardized residuals the model treats as stationary t_ν . Elevated values flag states in which the realized residual shape is heavier or more asymmetric than the fitted law typically produces. A correctly specified low- ν t_ν can itself generate high realized kurtosis, which is why the score is treated as an indicator of stress and makes no claim to be a specification test (Section 2.5). The data reject the finer per-name normalization. A plausible reason is that a name’s fitted ν is estimated from the same residuals and a low value is itself a symptom of the tail activity that matters, so dividing it out removes part of the signal.

Scale versus shape. The top decile follows large standardized residuals, typically price jumps, whose tails in individual stocks are heavy (Bollerslev et al., 2013), and since a GARCH(1,1) inflates its conditional variance after such a shock, part of the edge reflects the standard filter’s post-outlier scale error rather than any error in the innovation shape. The two can be separated with a jump-robust GARCH, which bounds the squared-innovation news term at $K\sigma_{t-1}^2$ (a $\sqrt{K}$ -sigma cap that limits how far a single shock propagates into the variance) and so responds to a single shock only within that cap. Against that benchmark the pooled body’s top-decile edge falls from +2.98% (DM 10.5) to +0.45% (DM 4.7) at a three-sigma cap and to +0.27% (DM 2.5) at four sigma. The robust filter alone beats standard GARCH by +3.17% (DM 8.2) in that decile, more than the estimator’s +2.98%, and the estimator rebuilt on the robust filter beats standard GARCH by +3.51% (DM 8.4) in the top decile of the robust-filter score (Table 2). Most of the frontier is therefore daily-scale misspecification, which a robust scale corrects on its own; what remains under a robust scale is a conditional-shape edge of the order of a third to a half of a percent, smaller but still significant. The large-cap realized-variance decomposition of Section 5 tells the same story. It is the reason the score is read as a monitor of parametric-model stress, part scale and part shape, and not as a diagnostic of shape alone. A Beta- t -EGARCH fitted by maximum likelihood (Harvey, 2013), whose score-driven update bounds the response to a large residual, makes the same point without a cap to choose. Under the frozen fits it beats GARCH- t by +3.56% (DM 8.8) in the top decile and +0.74% (DM 9.4) overall, and it beats the estimator on the GARCH- t scale in every region (+1.16%, DM 4.7, in the top decile; +0.48%, DM 5.1, overall). With the estimator rebuilt on the Beta- t -EGARCH scale, the flexible tail’s edge over that model’s own Student- t tail is +0.27% (DM 2.0) in the top decile and within noise elsewhere, so on a robust scale the frontier is flat. Against plain Beta- t -EGARCH, the estimator on its scale still has lower FZ0 at both levels under the frozen fits (DM 3.85 and 4.68) and under the annual refits (DM 3.09 and 2.72), and lower pinball loss at the 1% level under both (DM 5.4 and 2.1) and at the 2.5% level under the frozen fits (DM 3.4; 0.8 under refits). Averaged over the eleven pinball levels the two are within noise, because at the central levels they tie.

Two checks on the decomposition. Neither the learner’s inputs nor a richer parametric family accounts for the frontier. Retraining the Table 1 learner without its trailing residual-dispersion features leaves a top-decile edge of +1.93% (DM 10.4), and with the GARCH scale as its only

Table 2: Scale versus shape: pinball edge over standard GARCH- t (% , per-date DM in parentheses), 200 CRSP names, 221.6k asset-days, three-sigma news cap.

Scale	Shape	
	GARCH- t	Flexible (pooled body)
<i>All asset-days</i>		
Standard GARCH(1,1)	benchmark	+0.35 (5.35)
Jump-robust GARCH(1,1)	+0.42 (8.47)	+0.52 (7.45)
<i>Top decile of the score</i>		
Flexible over GARCH- t shape, standard scale		+2.98 (10.48)
Flexible over GARCH- t shape, robust scale		+0.45 (4.66)
Robust over standard scale, GARCH- t shape		+3.17 (8.20)
Robust scale and flexible shape over standard GARCH- t		+3.51 (8.44)

Edge = (reference pinball – model pinball)/reference pinball over the eleven-level grid; the reference is standard GARCH(1,1)- t except in the second top-decile row, where it is the jump-robust GARCH- t . The jump-robust filter uses the standard QMLE parameters and replaces the news term e_{t-1}^2 by $\min(e_{t-1}^2, 9\sigma_{t-1}^2)$. Top-decile rows sort on the score computed from the residuals of the filter in the comparison: the first and third on the standard filter, the second and fourth on the robust filter. With a four-sigma cap the five robust-filter entries become, in table order, +0.45 (10.30), +0.48 (6.71), +0.27 (2.47), +3.34 (8.92) and +3.55 (8.63).

input +1.63% (DM 11.0), with the decile ordering unchanged in both; the score is never a learner input, so about half of the top-decile edge is available from the scale state alone and the dispersion features add roughly a third (Online Appendix Table OA.14). A GARCH- t whose degrees of freedom move with the lagged absolute residual, fitted per name by maximum likelihood, gains +0.27% (DM 1.9) over constant- ν GARCH- t in the top decile, and the learner keeps +2.72% (DM 9.7) over it there (Online Appendix Table OA.15), so a state-adaptive shape within the parametric family does not reach the frontier.

Robustness of the frontier. Four checks in the Online Appendix address the main potential artifacts. The one-day score lag is not a look-ahead window: it gives the largest top-decile edge, and a five-day-old score keeps most of it (Online Appendix Figure OA.1). Re-forming the decile cutoff from strictly earlier dates, the rule the monitor uses, leaves 86% of top-decile membership unchanged and the edge intact (+2.45%, DM 8.3), so the concentration is not an artifact of the pooled-panel sort. A per-name FHS reweighted toward days of similar mk_{63} is significantly worse than plain FHS (per-date DM –11.3), so this reweighted-FHS specification does not absorb the edge and the pooled learner is doing the work. A double sort on mk_{63} and trailing volatility shows the edge needs both a shape break and an active regime, peaking at +2.33% where both are present (Online Appendix Table OA.2). A superior-predictive-ability (SPA) test (Hansen, 2005) against the industry benchmarks returns $T^{\text{SPA}} = 0$, and with the design-era threshold held fixed the top-decile edge is positive in every test year from 2020 to 2024. On simulated panels whose true model is known, the frontier appears only against a misspecified model (Online Appendix Table OA.16). When the data come from GARCH- t , the estimator on the GARCH- t scale trails the true model slightly (pinball 0.02%, DM –5.2; –1.9 in the top decile), while on a misspecified

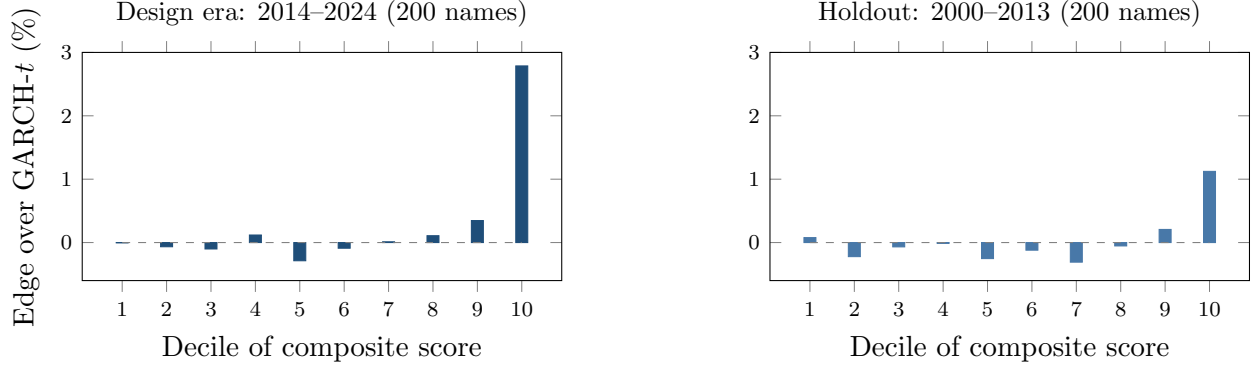


Figure 1: The frontier in and out of era, by decile of the composite misspecification score. Left: the design era (2014–2024). The top decile carries +2.79% at per-date DM 8.9, while deciles one through nine sit within $\pm 0.35\%$. Right: the composite score run once on 2000–2013 CRSP data. The top decile carries +1.12% at per-date DM 2.51 over 1,465 dates. The threshold shape (edge concentrated in the top decile, near zero elsewhere) reappears, as a post-hoc check, in an era containing the 2008 crisis that no design stage used.

Beta- t -EGARCH scale it beats that scale’s own Student- t tail (DM 10.2, and 7.7 in the top decile), and data generated from Beta- t -EGARCH give the mirror image.

The untouched-era holdout. Each design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze a simpler returns-space pipeline in full (signals, features, hyperparameters, split rule, and test statistics) and wrote down three predictions in advance: a positive and significant top-decile edge with a bottom decile near zero, a small positive overall edge, and a threshold-shaped decile profile matching Table 1. We then pulled CRSP daily returns for 2000–2013, an era not used at any earlier stage of the analysis. The pre-specified kurtosis-component frontier, the variant frozen in advance, replicates: the top-decile edge is +0.89% with per-date DM 2.16 ($p = 0.016$, 1,465 dates), deciles one through nine sit between -0.18% and $+0.28\%$, and the overall edge is +0.02%. The residual-hybrid composite replicates as well (Figure 1): its top decile carries +1.12% at per-date DM 2.51. This is the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. On the same holdout rows the pooled GARCH-EVT again shows no frontier (-0.15% against GARCH- t in the top decile, DM -2.8), and the frozen learner beats it there by +1.16% (DM 2.5), within noise elsewhere. The decile boundaries above are within-sample ranks of the holdout era, a partition that never touches a forecast but could not have been applied on day one. Two day-one-applicable rules recover the result: freezing the design-era thresholds as constants keeps the threshold shape (bins eight to ten at +0.82 to +1.02%, though the cell thins and the test loses power), and an expanding pooled 90th-percentile threshold over strictly earlier dates recovers the sort’s edge with significance (+0.99%, DM 2.15). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge, and the frozen thresholds carry +3.17% at DM 7.0 on the point-in-time universe of Section 3.2.

The overall signal–edge correlation is low (0.05–0.10) because the relationship is a threshold nonlinearity, so the top-decile Diebold–Mariano statistics are the appropriate summary.

Table 3: The frontier across universes. Ratio = nonparam/GARCH- t pinball (< 1 : non-parametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM -13.7
	EM (10)	1.013	8	GARCH, pooled DM -9.5
	crisis (7)	0.973	~ 1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86 , $p = .002$
	USDEGP	0.857	1920	DM 3.80 , $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins ^a
Cross-asset (43)	vol indices	~ 0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
	13/43 significant wins overall, corr(log kurt, edge) = 0.43			
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM -2.5 to -5

^a Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Exploratory: only Kenya survives the 93-comparison familywise adjustment (Online Appendix Table OA.4). Cross-index corr(log residual kurtosis, edge) = 0.53 . Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating that kurtosis is neither necessary nor sufficient.

4.2 The frontier across universes

Table 3 shows the same axis at work across markets. The largest standalone wins are the hyperinflation and capital-control currencies, the Argentine peso and the Egyptian pound, 12–14% better and individually significant, whose residual kurtosis (~ 2000 – 3300) sits two orders of magnitude above anything GARCH- t was built to absorb. Twenty-six country indices trace a gradient that tracks residual kurtosis more closely than the development label, and forty-three cross-asset instruments extend the pattern into rates, commodities, and credit. The Korea-2026 crash is the negative control: a price crash under circuit breakers censors the return but leaves residual kurtosis ordinary, and GARCH- t correctly wins all twenty-three assets, so the frontier tracks residual shape and not price turbulence. The instrument-level entries are exploratory, and under a Holm step-down over the 93 comparisons (Online Appendix Table OA.4) seven survive. The aggregate crisis-tier FX statistic (DM 4.12) and the cross-universe kurtosis–edge gradients are the evidence this section rests on. The five daily crypto series show no edge; a frequency dimension is left to future work.

4.3 Regime dependence and model selection

In the amortized panel the absolute edge grows about fourfold with turbulence, from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent, and both nonparametric estimators beat GARCH in every quintile. Gating, running GARCH on calm days and the nonparametric model when the score is high, does not improve on continuous use of the flexible model: a classifier gate routes 90.5% of days to the nonparametric model and matches always-nonparametric exactly, and a regression gate on the edge magnitude (Giacomini and White, 2006) still converges to it.

This is consistent with Table 1: the conditional mean of the loss differential is positive where the score is high and indistinguishable from zero elsewhere, so no state pays to select the parametric quantile, and the per-day oracle gap (4.4%) is largely unforecastable from prior-day state. The amortized setting therefore uses the nonparametric model throughout; the residual-hybrid is the choice where forecast accuracy no worse than GARCH- t in the bulk is wanted, with the score as a monitor (Section 6).

5 VaR and Expected Shortfall forecast evaluation

5.1 Head-to-head accuracy and significance

Table 4 reports the full comparison on the 200 CRSP names of Table 1 (221.6k asset-days, 1,108 dates).

The stress-era replication. The comparison set was re-run on the pre-committed 2000–2013 holdout panel. Splits follow Section 3, so each name’s test window runs from roughly mid-2008 through 2013, the crisis and its aftermath. The EVT-tailed residual-hybrid passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t -statistics of 0.17 at 99% and 0.35 at 97.5%, consistent with nominal at both regulatory levels through the crisis window. Historical simulation and EWMA do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH- t is also well calibrated in this era (94% pass rate), as the frontier predicts. Calibration parity is the norm, and the residual-hybrid’s added value in any era is the accuracy edge where the misspecification score is high. The adaptive overlay of Stage 4 stays near nominal through the crisis window, breaching at 2.43% and 2.38% against the 2.5% target and at 1.06% and 1.04% against 1% for the two update rates (date-clustered $|t| \leq 0.44$).

Non-core comparisons. Table 5 adds two entrants from outside banks’ standard toolkit. SAV-CAViaR, a leading semiparametric academic benchmark, ties the residual-hybrid, and the 90% Model Confidence Set on that run contains exactly these two, so the residual-hybrid ties the strongest non-nested academic rival and significantly beats every model in banks’ standard toolkit. The two tied models differ in what they deliver: SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history, so it cannot be applied to new listings; the residual-hybrid produces the full curve and extends to new listings. Pooled exception rejections are not counted against CAViaR, because pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, the residual-hybrid included; the date-clustered restatement is the appropriate reading. A neural IQN used as the residual model loses to trees at both tunings (Table 5): the tail-aware, monotone, conformally recalibrated version narrows the untuned gap by an order of magnitude and repairs the thin tail, but the remaining deficit is still significant, so trees are the stronger tabular estimator and the residual-hybrid’s edge does not rest on the choice of shape learner. Two textbook FHS variants, per-name train-constant and per-name rolling 500-day trailing, were run on the identical panel;

Table 4: FRTB comparison: average pinball loss (twelve quantile levels), ES_{97.5} calibration, and 99% exception tests. 200 CRSP names, 221.6k test asset-days.

Model	Avg. pinball	ES _{97.5} pred. vs realized	Breach ₉₉	Kupiec ₉₉ p
<i>Ideal value</i>	<i>(smaller better)</i>	<i>(pred. = realized)</i>	<i>1.00%</i>	<i>> 0.05 (pass)</i>
Residual-hybrid (GBM)	0.3676	−6.64 / −6.12	1.06%	0.01
+ EVT tail	0.3680	−7.33 / −6.16	0.87%	0.00
GARCH(1,1)- t	0.3688	−6.98 / −5.92	1.03%	0.167
GJR-GARCH-skew- t	0.3688	−6.87 / −6.09	1.02%	0.431
GARCH-FHS (per-name)	0.3689	−6.90 / −6.06	1.08%	0.00
GARCH-FHS (pooled)	0.3691	−7.17 / −5.89	0.97%	0.206
GARCH-FHS (rolling 500d)	0.3694	−7.02 / −6.21	1.10%	0.00
EWMA (RiskMetrics)	0.3735	−5.72 / −5.84	1.61%	0.00
Historical simulation	0.3752	−7.71 / −7.08	0.87%	0.00

All columns come from a single common-sample run, provided in the replication package, on the 200 names and 1,108 test dates of Table 1 (221.6k asset-days), so the battery, the frontier and the FZ0 comparison share one set of rows. Predicted ES is the tail integral $\alpha^{-1} \int_0^\alpha Q(v) dv$ of each model’s quantile function: closed forms for the t and normal entrants, an exact substitution integral of the Hansen skew- t inverse CDF, the exact empirical tail mean for historical simulation and the FHS variants, and the converged integral of Section 2.2 for the EVT-tailed residual-hybrid, whose VaR and ES are both read from one monotonized min-envelope curve Q_t^z of (5). The raw residual-hybrid has no quantile below its lowest fitted level, so its ES extends the body flat below $\alpha/40$ and is a lower bound on its severity. Diebold–Mariano vs the residual-hybrid: GARCH- t 6.2, GJR-skew- t 5.6, FHS pooled 6.9, per-name 6.3, rolling 6.0, HS 8.5, EWMA 9.6 (all $p \approx 0$, rounded, and stable across reruns). The EVT variant concedes 0.0004 pinball (DM 7.0) to the first row, the raw residual-hybrid without the tail, as the price of that tail. The 90% MCS contains the raw residual-hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a diagnostic, not a ranking: “realized” is the mean return below each model’s own VaR, so the conditioning set is model-dependent, and under the numerical integral each fat-tailed entrant exceeds its own-tail comparator by 8% (raw residual-hybrid, a lower bound) to 22% (pooled FHS). The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew- t row: Hansen inverse CDF with the fitted degrees of freedom and skewness (median 4.2 and -0.015), validated against the symmetric- t limit and against an independent reference implementation. Exception columns are pooled panel diagnostics: at 221.6k observations a breach-rate deviation of ± 0.1 points rejects in either direction (the EVT variant’s 0.87% sits on the conservative side; GARCH- t , GJR-skew- t and pooled FHS land within 0.03 points of 1%), so pooled p -values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement: per asset the EVT-tailed residual-hybrid of this table passes Kupiec at 99% for 86% of names (Online Appendix Figure OA.2), and the accuracy layer of the FZ0 comparison passes Kupiec for 84% and Christoffersen conditional coverage for 90% (at 97.5%, 70% and 75%). On the date-clustered test it passes at 99% ($t = 0.62$) and is rejected marginally at 97.5% ($t = 2.00$), where GARCH- t passes ($t = 1.56$) and pooled FHS is rejected more strongly ($t = 3.35$); its pooled body without the EVT tail is rejected at both levels ($t = 4.3$ and 4.7), so the tail stage carries the regulatory calibration. All three pass on the 2000–2013 holdout, and the adaptive overlay of Stage 4 passes on both panels.

both lose to the residual-hybrid (DM 5.2 and 5.7, pinball 0.3562 and 0.3569 against 0.3551) and bracket the pooled variant used in Table 4.

The joint (VaR, ES) score. Because ES is not elicitable alone but the pair (VaR, ES) is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the comparison set was also

Table 5: Non-core comparisons. Each entrant is scored against the residual-hybrid on its own common sample, so loss levels are comparable within a row but not across rows.

Entrant	Loss	Residual-hybrid	DM	Breach ₉₉	Verdict
SAV-CAViaR	0.3461	0.3460	0.4	—	tie ($p = 0.34$)
Neural IQN, untuned	0.3650	0.3551	13.3	3.2%	loses
Neural IQN, tuned ^a	0.3638	0.3632	3.03	0.92%	loses

^a Tail-aware τ -sampling, monotone rearrangement, and a per-level conformal shift calibrated on the first half of the test window and scored on the second half; the residual-hybrid in this row is recalibrated the same way, so the loss levels are not comparable with Table 4.

DM is the per-date Diebold–Mariano t -statistic against the residual-hybrid; positive favors the residual-hybrid. The 90% Model Confidence Set contains the residual-hybrid alone on the Table 4 run, and the residual-hybrid together with CAViaR on the CAViaR run. Breach rates were not computed for the CAViaR run, which reports one quantile level per fit.

scored under the FZ0 joint score

$$L_{\text{FZ0},\alpha}(r, q, e) = -\frac{1}{\alpha e} \mathbb{I}\{r \leq q\}(q - r) + \frac{q}{e} + \log(-e) - 1, \quad e \leq q < 0, \quad (10)$$

the 0-homogeneous member of the class (Patton et al., 2019). Lower values are better. On the large-cap sandbox subset used to fix the score’s orientation, a Gaussian-innovation GARCH is significantly worst at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, Gaussian innovations are clearly inadequate. Among the fat-tailed entrants the FZ0 differences on large caps are small ($|\text{DM}| < 1.4$), the frontier’s own prediction that the shape edge is flat on well-behaved names. On a high-kurtosis subset the ordering favors the shape-aware forecaster, directionally but not significantly at that width. The pre-specified full-panel run, reported next, settles the magnitudes on the residual-hybrid’s own forecasts.

On the full 200-name panel (221.6k test asset-days), VaR and ES are read from the coherent min-envelope curve Q_t^z of (5) and per-date DM inference is used throughout. The residual-hybrid’s *accuracy layer*, the EVT-tailed variant with no conformal shift, attains the lowest FZ0 of the industry-standard benchmarks of Section 1 (historical simulation, EWMA, GARCH- t , GJR-GARCH-skew- t and FHS) at 1% and is never significantly beaten by them at 2.5%. At 1% it beats GARCH- t (DM 4.9) and FHS (DM 4.9), the two standard benchmarks the accuracy claim rests on. At 2.5% it again wins over GARCH- t (DM 5.5). On the same rows it also beats standalone GARCH-EVT, whose ES is the McNeil–Frey closed form, with a pooled tail by DM 4.5 at 1% and 6.3 at 2.5%, and per name by DM 5.1 and 5.4, so the gain over the conventional EVT construction is not confined to the pinball frontier. The dynamic quantile test of Engle and Manganelli (2004) (four hit lags and the VaR) passes at the 5% level for 86% of names at 1% and 70% at 2.5% for the accuracy layer, against 84% and 68% for GARCH- t and 83% and 62% for the pooled GARCH-EVT. Because a ranking under one consistent score need not hold under another (Patton, 2020), the Online Appendix reports Murphy diagrams (Ehm et al., 2016) for the VaR forecasts: the accuracy layer has the lower elementary score against GARCH- t on 70% of the threshold grid at 1% and 65%

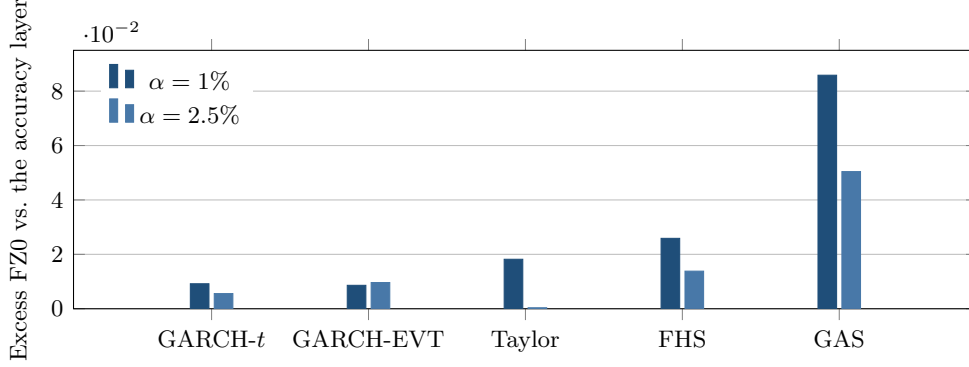


Figure 2: Joint (VaR, ES) FZ0 loss of each benchmark in excess of the residual-hybrid’s accuracy layer (the EVT-tailed variant without the conformal shift), on the full 200-name panel. A positive bar means worse than the accuracy layer. The accuracy layer (the zero baseline) attains the lowest FZ0 at 1% and is never significantly beaten at 2.5%, beating GARCH-*t* (DM 4.9 at 1%, 5.5 at 2.5%), pooled-tail GARCH-EVT (DM 4.5 and 6.3) and per-name FHS (DM 4.9 and 5.4), together with the two FZ-estimated dynamic-ES benchmarks fitted per name by multi-start FZ0 minimization, the generalized autoregressive score (GAS) model of Patton et al. (2019) (DM 9.9 and 8.8) and the ES-CAViaR of Taylor (2019) (DM 3.8 at 1%, a tie at 2.5%). Every bar and DM comes from one run on the same rows against the same accuracy-layer forecast. SAV-CAViaR, scored on its own common sample (Table 5), ties the accuracy layer and is omitted here; the adaptive overlay, within noise of the accuracy layer at both levels, is discussed in the text.

at 2.5%, and against the pooled GARCH-EVT on 85% at both levels, losing in a band of thresholds near the typical VaR level. Extending the joint score to the whole comparison set on the same rows (Online Appendix Table OA.11), the accuracy layer has the lowest mean FZ0 of the benchmark models at 1%, a level statement since GJR-GARCH-skew-*t* is within noise (DM 1.2). At 2.5% the accuracy layer, GJR-GARCH-skew-*t* and the Taylor model form a tie group (FZ0 1.8486, 1.8486 and 1.8490; DM -0.02 and 0.1), and nothing significantly beats the accuracy layer; its own pooled body, without the EVT branch, scores lower still (DM -1.1 at 1% and -1.5 at 2.5%, both within noise) while breaching at 1.3% and 3.3%, the coverage cost the EVT branch removes; the 90% Model Confidence Set on the per-date FZ0 series contains the accuracy layer, the pooled body and GJR-GARCH-skew-*t* at 1% (with the static-shift row, identical to the accuracy layer at that level) and the accuracy layer, the pooled body, GJR-GARCH-skew-*t* and the Taylor model at 2.5%, and excludes GARCH-*t*, every FHS variant, both GARCH-EVT forms, the GAS model, EWMA and historical simulation at both levels. On the eleven-level pinball the set contains the pooled body and SAV-CAViaR and excludes the accuracy layer, since the EVT branch costs pinball, as Table 4 reports for the twelve-level battery, and is kept for the exception tests it passes. The two closest dynamic (VaR, ES) benchmarks are the one-factor score-driven GAS of Patton et al. (2019) and the ES-CAViaR of Taylor (2019). Each is estimated per name on the same panel by minimizing the same FZ0 loss, and under a three-start optimizer both converge for all 200 names. The accuracy layer beats the GAS at both levels (DM 9.9 at 1%, 8.8 at 2.5%), and it beats the Taylor model at 1% (DM 3.8) while tying it at 2.5%, the near-tie the fat-tailed benchmarks all show at 2.5% where the shape edge is thin. A score-driven Beta-*t*-EGARCH (Harvey, 2013), outside the industry-standard set, has lower FZ0 at both levels under the frozen fits, significantly at 2.5% (DM 2.48). Under the annual refits of Section 3.1 the two are within noise overall, with the accuracy layer slightly ahead

(DM 1.2 and 0.9), while Beta- t -EGARCH stays significantly ahead in the top decile of the score (DM 3.7 and 5.1). Its bounded response to a shock removes the post-shock variance overstatement of Section 4, the frontier’s own mechanism, and refitting does not.

This ranking survives the annual walk-forward refit of Section 3.1: re-estimating every stage at each January-1 cutoff and re-scoring the 2020–2024 panel, the accuracy layer keeps the lowest FZ0 at both levels, beating GARCH- t by DM 3.5 at 1% and 3.1 at 2.5%, and pooled FHS, the unconditional residual quantile on the same refitted scale, by DM 3.2 and 3.4 (one-sided $p < 0.005$). Refitting narrows these margins without closing them, unlike the average pinball edge, which closes under the same refit. With Beta- t -EGARCH in place of GARCH- t as Stage 1, the engine has lower FZ0 under the frozen fits (DM 3.16 and 4.37, and 3.14 and 3.63 on the 2000–2013 holdout). Under annual refits the two engines are within noise overall (DM -0.31 and -0.02), and the Beta- t -EGARCH engine is ahead in the top decile of the score (DM 5.46 and 7.37), so refitting does on average what a robust filter does but leaves the post-shock error in the states the score flags. Stage 1 stays GARCH- t under the rule fixed before the refit run.

The accuracy layer is thus one configuration that is never significantly beaten on FZ0 by the industry-standard benchmarks at either level, scores lowest at 1%, and passes the aggregate date-clustered exception test at 99%. At 97.5% it is rejected marginally on 2020–2024 ($t = 2.00$, against 1.56 for GARCH- t and 3.35 for pooled FHS) and passes on the 2000–2013 holdout ($t = 0.25$), a test with little power there: its clustered standard error for the accuracy layer is 0.53 points at 1% and 0.28 at 2.5%, so it rejects no breach rate below 2.05% at 1% or inside 1.95–3.05% at 2.5%. Per-name diagnostics show residual cross-sectional calibration heterogeneity (84% and 90% pass rates at 99%). The pooled 221.6k-observation Kupiec test also rejects at 97.5%, and at this sample size it rejects very small deviations. The joint-score edge over GARCH- t is less concentrated than the pinball edge: at 1% the mean FZ0 gap is 0.011 in the top misspecification decile and 0.009 in the other nine (DM 3.4 and 4.7). That flatness is a property of the splice. Dropping the EVT branch and scoring the pooled body alone, the body beats the accuracy layer in the top decile by 0.068 at 1% (DM -6.4) and 0.060 at 2.5% (DM -6.8), and ties it in the other nine deciles (0.002 and 0.001, DM 0.4 and 0.2). The pooled GPD is unconditional, so it sits at a fixed level in standardized-residual terms, -2.55 at 1% and -1.88 at 2.5%, while the body pulls its own tail quantile in from -2.53 to -2.25 across the deciles as the score rises, and the minimum of the two therefore overrides the body by a margin that widens from 0.19 to 0.52. Anchoring the same GPD to each row’s own body quantile at the splice level, under a rule fixed before the run, removes the top-decile deficit (DM $+0.1$ against -6.6 for the unconditional tail at the same splice level) and raises the 97.5% breach rate from 2.78% to 3.09%. The unconditional tail therefore buys about 0.3 points of 97.5% coverage at a cost of 0.06 to 0.07 in top-decile FZ0 and 0.013 to 0.019 overall, and the paper keeps it for the coverage.

The adaptive overlay. Stage 4 shifts the accuracy layer by a scalar updated each day from the panel breach frequency (Section 2.2). Before the run we fixed the rule under which it would be switched on: on both the 2014–2024 panel and the 2000–2013 holdout, at both update rates, the overlay had to bring the 97.5% breach rate within 0.3 points of nominal and keep its FZ0 within

noise of the accuracy layer ($|\text{DM}| < 2$) at both levels. It meets both conditions. In era it moves the 97.5% breach rate from 2.78% to 2.68% and 2.60% for $\gamma = 0.02$ and 0.05, with FZ0 slightly below the accuracy layer’s (DM -1.3 and -1.0), and the date-clustered test passes ($t = 1.30$ and 0.78); at 99% it leaves the accuracy layer essentially unchanged (DM -0.3 for both rates). On the holdout its breach rates are 2.43% and 2.38% at 97.5% and 1.06% and 1.04% at 99%, with every $|\text{DM}|$ against the accuracy layer below 1. It keeps the joint-score lead over GARCH- t at 2.5% in era (DM 5.2 and 4.1). In the top misspecification decile it concedes 2.5% FZ0 to the accuracy layer (DM 3.4 and 3.6), the price of widening the band where it is already widest. A static shift, the construction of earlier versions, fails in both of its forms. Estimated on the body’s calibration errors and applied to the envelope, it counts the tail correction twice and over-covers (breach 1.66%, date-clustered $t = -9.1$, FZ0 DM 2.6 against the accuracy layer); estimated on the envelope’s own errors, it still over-covers in era (2.15%, $t = -3.1$), because the calibration block and the test era miss on opposite sides of nominal. Each “lowest FZ0” statement refers to the accuracy layer.

Direct ES calibration backtests. Because exception counts test VaR and not ES, two direct ES calibration backtests, the exceedance-residual test of McNeil and Frey (2000) and the Z_2 statistic of Acerbi and Székely (2014), complement the jointly consistent FZ0 loss above. Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a joint score. Both are computed on the residual-hybrid’s own min-envelope forecasts over the same 200-name panel, with a stationary block bootstrap over the 1,108 test dates (Online Appendix). The Gaussian filter is worst on each ES metric, and among the fat-tailed entrants no single model dominates each diagnostic. The residual-hybrid has the Z_2 closest to zero at 1% and ties GARCH- t at 2.5%, and the residual-hybrid and GARCH- t are the two models that pass the 1% Kupiec test. On the exceedance residual the residual-hybrid is closest to zero at 1% and FHS, which targets the empirical tail directly, at 2.5%. Neither test rejects the residual-hybrid’s ES at either level (exceedance residual $p = 0.38$ and 0.58 , Online Appendix Table OA.5), nor under the annual refits ($p = 0.21$ and 0.33), while GARCH- t ’s 1% ES is rejected as not severe enough ($p < 0.001$). On the 2000–2013 holdout the verdict depends on the direction of the error. At 1% the test rejects every construction, GARCH- t ’s ES as not severe enough (mean exceedance residual -0.17) and the residual-hybrid’s and FHS’s as too severe ($+0.15$ and $+0.12$); at 2.5% the residual-hybrid passes ($p = 0.48$) and GARCH- t is rejected ($p = 0.02$). These verdicts require an accurate ES: with the 20-node midpoint ES of earlier versions, the exceedance-residual test rejects a correctly specified model in 22.5% of simulated samples at a nominal 5% (Online Appendix Table OA.16). The comparative result is the FZ0 loss, where the edge over GARCH- t and FHS is significant at both levels.

5.2 The ten-day horizon: a direct multi-day extension

The multi-day results come from a direct amortized model trained on h -day cumulative returns; the residual-hybrid of Algorithm 1 is a one-day architecture, and no ten-day claim attaches to it. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) its pinball edge over $\sqrt{h}$ -scaled GARCH grows from $\sim 0.4\%$ at $h = 1$ to $\sim 1.4\%$ at $h = 10$ and holds in the 2020–2022 stress window (Online

Appendix Figure OA.3). About half of that edge is the scaling rule itself. The $\sqrt{h}$ rule holds the one-day variance flat across the window, while the fitted GARCH variance mean-reverts toward an unconditional level that sits 6–12% above h times the one-day variance in every year of the 2020–2024 test window; against a GARCH- t whose variance is iterated through the window the direct model’s edge is 0.8% (DM 4.0), against simulated GARCH- t paths 0.7% (DM 3.5), and against bootstrapped filtered paths 0.5% (DM 2.5, and 1.1 on non-overlapping dates), on the same rows. The ten-day tail advantage is real in that era but does not survive the pre-committed 2000–2013 rerun, where the fitted term structure is flat (iterated ten-day variance within 2% of h times the one-day variance in 2008–2009): out of era the two methods tie on accuracy (direct edge +0.52%, DM 0.09); against the iterated benchmarks the point estimate is the same 0.5–0.8% as in era, but with DM at most 0.3, because per-date loss differences through 2008 are far noisier; and the calibration verdict reverses, with $\sqrt{h}$ -scaled GARCH- t almost exactly calibrated through the crisis while the direct tail under-states it. A level-wise minimum of the two curves does not repair this. In 2008–2009 it breaches at 1.61% and 3.52% against the 1% and 2.5% targets, closer to nominal than $\sqrt{h}$ scaling (1.93% and 3.99%) but short of it, and in era it costs 1.1% of pinball against the direct model (DM 3.9), so we do not recommend it. Rescaling the direct curve by the fitted variance term structure ties the direct model in era (DM -1.7 ; -0.3 on non-overlapping dates) with breach rates nearer nominal (0.97% and 2.44%) and loses to it out of era (DM 2.3); we record it as exploratory. The ten-day evidence is therefore mixed; the benchmark table is Online Appendix Table OA.13.

5.3 Cross-asset evidence

Across the 43-instrument sweep the relationship is heterogeneous. The volatility indices (VIX, V2X, VVIX) and Baltic freight post the largest accuracy gains and survive the multiplicity adjustment (Online Appendix Table OA.4), but their calibration is open. VIX’s breach-independence is marginal (one-sided $p = 0.041$), and credit and Baltic freight fail it, so all are reported as accuracy edges and none as a calibrated win. Instrument-level results and diagnostics are collected in the Online Appendix.

The scale–shape decomposition. Against a Realized GARCH (Hansen et al., 2012) fitted on intraday data, the daily-data core concedes FZ0 at both levels (DM 6.2 and 4.6). Re-fitting the shape learner, EVT tail, and score on the realized residuals, however, leaves the score’s edge within noise. On large caps the frontier is therefore largely daily-scale misspecification. Where intraday data are unavailable or too sparse to build a realized measure (new listings, illiquid names, many cross-asset instruments), the daily score is the version of the diagnostic that can be computed (Online Appendix).

6 Extensions and implications

Using the score. The score is available *ex ante*: a risk manager computes today’s value from today’s residuals and uses it for tomorrow’s forecast. Its role is to stratify, not to switch models day by day, because the per-day oracle gap (4.4%) is largely unforecastable from prior-day state and

a learned gate collapses to always-nonparametric (Section 4.3). It drives a switch in two settings: the per-name standalone model, where nonparametric quantiles underperform on calm days, and the choice of Stage-1 scale, described next. As a monitor it is cheap, since its input is the trailing residual window the scale stage already produces, and it is a slow state that need not be watched daily. On capital, the residual-hybrid is a one-day single-name VaR/ES forecaster of the kind that feeds an internal model, not the ten-day portfolio-level stressed ES that sets the FRTB charge; the paper makes no capital-release claim, because the predicted-versus-realized ES wedge conditions on each model’s own breach set and no defensible arithmetic converts it to capital (Basel Committee on Banking Supervision, 2019). The narrower implication is that the ES-based component of the charge scales with the ES forecast.

Choosing the scale with the score. Under the annual refits the Beta- t -EGARCH engine and the GARCH- t engine are within noise overall while the robust one leads in the top decile of the score (Section 5), which suggests letting the score choose the scale. A gate that runs the GARCH- t engine when the previous day’s mk_{63} is below its pre-2020 90th percentile, a threshold fixed before the first test year, and the Beta- t -EGARCH engine otherwise switches on 10.0% of the 251.6k refit test asset-days. In a run whose predictions were written beforehand, its joint score is significantly better than always running the GARCH- t engine (DM 6.3 at 1%, 8.1 at 2.5%) and statistically equal to always running the Beta- t -EGARCH engine (DM 0.8 at both levels), which is slightly ahead of it in the top decile (DM -1.6 and -1.7 , within noise); the gate and both fixed engines pass the date-clustered exception test at both levels. The gate keeps the paper’s Stage-1 filter on nine asset-days in ten and recovers the robust scale’s gain where the score is high. Where tail accuracy is the only objective, running the robust scale everywhere does as well.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant here that can produce a conditional risk forecast for an asset with no return history. An IPO or newly listed ETF gets day-one quantiles from its characteristics and first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names; the residual-hybrid of Section 2.2 attaches once a scale filter is estimable. Day-one quantiles exist but are not calibrated at the regulatory levels. In a separate run on 303 names listed from 2018, the best pooled peer forecast breaches its 1% VaR on 7.1% of asset-days in the first five trading days. Once the name has a few weeks of history the amortized forecast beats its own-history benchmark, an expanding-window Gaussian scored at seven levels from 0.05 to 0.95, by about 9% of pinball at 15 to 30 trading days and 6.5–7% through the first year, and on 91.7% of test dates overall. The ablation in Section 2.3 traces the handover from characteristics to the name’s own realized volatility as its history lengthens.

7 Conclusion

A fixed-shape parametric filter is hard to beat where its shape assumption holds, and for most assets on most days the flexible model does not measurably improve on it. The flexible model’s advantage

is concentrated in a small set of high-stress states identified by the real-time misspecification score. The decomposition attributes most of this advantage to post-outlier variance overstatement by the standard GARCH filter; a smaller but significant conditional-shape component remains under a jump-robust scale. An estimator that exploits both components, while showing no detectable accuracy loss relative to the benchmark elsewhere, improves on the accuracy of the market-risk models in standard use when judged by one-day diagnostics at the FRTB backtesting levels, at calibration parity with the best of them and short of the desk-level approval a full internal model must pass. Because the shape model is pooled across names, its characteristics-only amortized variant produces a forecast for a newly listed asset from its first day, though not yet a calibrated one in its first week, and the residual-hybrid attaches once a scale filter is estimable (Section 6).

These conclusions carry qualifications. The score is predictive rather than a necessary or sufficient condition: high residual kurtosis coincides with ties in some markets, and in credit and freight the accuracy edges sit alongside failed breach-independence tests. The accuracy layer’s 97.5% coverage is rejected marginally on 2020–2024 (date-clustered $t = 2.00$, against 1.56 for GARCH- t), which the adaptive overlay repairs at no cost in FZ0. The McNeil–Frey exceedance-residual test rejects none of the residual-hybrid’s ES forecasts on the design panel or under annual refits, but on the 2000–2013 holdout it rejects the 1% ES of every construction tried, the residual-hybrid’s and FHS’s as too severe and GARCH- t ’s as not severe enough. The shape component is the smaller part of the edge, since the decomposition attributes most of it to daily-scale misspecification that a jump-robust or realized-variance scale removes (Sections 4 and 5), and what survives is real but modest; on a robust scale it sits in the tail, where the flexible tail still lowers the joint (VaR, ES) loss at both regulatory levels under frozen fits and annual refits. Where intraday data are too sparse to build a realized measure, the daily model and its score remain the available tool. The ten-day extension is a separate direct model: it beats $\sqrt{h}$ -scaled GARCH on the 2014–2024 panel, with about half of that edge surviving an iterated GARCH benchmark; on the pre-committed 2000–2013 rerun the point estimate against the iterated benchmark is unchanged but within noise, and the scaled parametric tail is the better calibrated of the two (Section 5). And the score serves as a monitor rather than a day-to-day switch between parametric and flexible shapes, since the day-ahead oracle gap is largely unforecastable (Section 4.3). It can still choose the Stage-1 scale: a gate that moves to Beta- t -EGARCH in the score’s top decile is significantly better than GARCH- t at both levels and statistically equal to running the robust scale everywhere (Section 6). Distributional uncertainty around these VaR and ES forecasts and the multivariate portfolio tail are left to future work.

Funding

This work was supported by the Northwestern University Department of Statistics and Data Science [Undergraduate Research Fellowship Grant, 2026].

Acknowledgments

I thank Wenxin Jiang for his advice and guidance throughout this research, and seminar participants and colleagues for comments. All errors are mine. AI assistance (Anthropic’s Claude) was used in drafting, editing, and code development; the author designed the research, verified all results, and is responsible for the content.

Conflict of interest

The author declares no conflicts of interest.

Ethics statement

This study uses licensed secondary market data (CRSP/WRDS and Bloomberg) under the terms described in the data availability statement. It involves no human subjects and required no ethics approval.

Data and code availability

A public replication package at github.com/sean-qin-usa/downside-risk-paper covers every WRDS-based table and figure in this paper and preserves the Bloomberg-based exhibits as run from their derived series. It provides estimation code for all stages of the estimator ((2)–(5)), the misspecification score (9), the FRTB comparison, and the walk-forward schedule, seeds, and hyperparameter grids referenced in the text, together with a mapping from each table and figure to the script and result file that produce it. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author’s institutional licenses and cannot be redistributed; the package includes the code to rebuild each panel from those sources, the derived non-proprietary series, and a synthetic example dataset on which the full pipeline runs end-to-end without licensed data. The holdout’s frozen specification, including its written-in-advance predictions, is recorded in the package alongside the design-era pipeline so that the two can be compared line by line. A citable archival snapshot (Zenodo DOI) will be minted for the accepted version.

References

- Acerbi, C. and Székely, B. (2014). Back-testing expected shortfall. *Risk*, 27(11):76–81.
- Acerbi, C. and Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking and Finance*, 26(7):1487–1503.
- Andersen, T. G. and Bollerslev, T. (1998). Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905.

- Artzner, P., Delbaen, F., Eber, J.-M., and Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3):203–228.
- Balkema, A. A. and de Haan, L. (1974). Residual life time at great age. *Annals of Probability*, 2(5):792–804.
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2023). Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816–845.
- Barone-Adesi, G., Giannopoulos, K., and Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5):583–602.
- Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk. Technical report, Bank for International Settlements.
- Bayer, S. and Dimitriadis, T. (2022). Regression-based expected shortfall backtesting. *Journal of Financial Econometrics*, 20(3):437–471.
- Berkowitz, J. (2001). Testing density forecasts, with applications to risk management. *Journal of Business and Economic Statistics*, 19(4):465–474.
- Berkowitz, J. and O’Brien, J. (2002). How accurate are value-at-risk models at commercial banks? *Journal of Finance*, 57(3):1093–1111.
- Bissiri, P. G., Holmes, C. C., and Walker, S. G. (2016). A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3):542–547.
- Bollerslev, T., Hood, B., Huss, J., and Pedersen, L. H. (2018). Risk everywhere: Modeling and managing volatility. *Review of Financial Studies*, 31(7):2729–2773.
- Bollerslev, T., Todorov, V., and Li, S. Z. (2013). Jump tails, extreme dependencies, and the distribution of stock returns. *Journal of Econometrics*, 172(2):307–324.
- Boudt, K., Daníelsson, J., and Laurent, S. (2013). Robust forecasting of dynamic conditional correlation GARCH models. *International Journal of Forecasting*, 29(2):244–257.
- Bucci, A. (2020). Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531.
- Carnero, M. A., Peña, D., and Ruiz, E. (2007). Effects of outliers on the identification and estimation of GARCH models. *Journal of Time Series Analysis*, 28(4):471–497.

- Chernozhukov, V., Fernández-Val, I., and Galichon, A. (2010). Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125.
- Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2018). Exact and robust conformal inference methods for predictive machine learning with dependent data. In *Proceedings of the 31st Conference on Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 732–749.
- Christensen, K., Siggaard, M., and Veliyev, B. (2023). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4):841–862.
- Chronopoulos, I., Raftapostolos, A., and Kapetanios, G. (2024). Forecasting value-at-risk using deep neural network quantile regression. *Journal of Financial Econometrics*, 22(3):636–669.
- Dabney, W., Ostrovski, G., Silver, D., and Munos, R. (2018). Implicit quantile networks for distributional reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1096–1105.
- Diebold, F. X. (2015). Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of Diebold–Mariano tests. *Journal of Business and Economic Statistics*, 33(1):1–9.
- Diebold, F. X., Gunther, T. A., and Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4):863–883.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.
- Ehm, W., Gneiting, T., Jordan, A., and Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(3):505–562.
- Embrechts, P., Klüppelberg, C., and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- Engle, R. F. and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381.
- Fissler, T. and Ziegel, J. F. (2016). Higher order elicibility and Osband’s principle. *Annals of Statistics*, 44(4):1680–1707.
- Francq, C. and Zakoïan, J.-M. (2015). Risk-parameter estimation in volatility models. *Journal of Econometrics*, 184(1):158–173.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232.

- Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578.
- Gibbs, I. and Candès, E. (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 1660–1672.
- Glosten, L. R., Jagannathan, R., and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801.
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494):746–762.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Gneiting, T. and Ranjan, R. (2011). Comparing density forecasts using threshold- and quantile-weighted scoring rules. *Journal of Business and Economic Statistics*, 29(3):411–422.
- Grané, A. and Veiga, H. (2014). Outliers, GARCH-type models and risk measures: A comparison of several approaches. *Journal of Empirical Finance*, 26:26–40.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5):2223–2273.
- Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3):705–730.
- Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380.
- Hansen, P. R., Huang, Z., and Shek, H. H. (2012). Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6):877–906.
- Hansen, P. R. and Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7):873–889.
- Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2):453–497.
- Harvey, A. C. (2013). *Dynamic Models for Volatility and Heavy Tails: With Applications to Financial and Economic Time Series*. Number 52 in Econometric Society Monographs. Cambridge University Press, Cambridge.
- Hoga, Y. and Demetrescu, M. (2023). Monitoring value-at-risk and expected shortfall forecasts. *Management Science*, 69(5):2954–2971.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.

- Hull, J. and White, A. (1998). Incorporating volatility updating into the historical simulation method for value-at-risk. *Journal of Risk*, 1(1):5–19.
- Jiang, R., Hu, X., and Yu, K. (2022). Single-index expectile models for estimating conditional value at risk and expected shortfall. *Journal of Financial Econometrics*, 20(2):345–366.
- Jiang, W. and Tanner, M. A. (2008). Gibbs posterior for variable selection in high-dimensional classification and data mining. *Annals of Statistics*, 36(5):2207–2231.
- J.P. Morgan/Reuters (1996). RiskMetrics — technical document. Technical report, J.P. Morgan/Reuters, New York.
- Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.
- Kuester, K., Mittnik, S., and Paolella, M. S. (2006). Value-at-risk prediction: A comparison of alternative strategies. *Journal of Financial Econometrics*, 4(1):53–89.
- Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3):1217–1241.
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84.
- Lei, J., G’Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. (2018). Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- Luo, Y., Xue, X., and Izzeldin, M. (2025). When MIDAS meets LASSO: The power of low-frequency variables in forecasting value-at-risk and expected shortfall. *Journal of Financial Econometrics*, 23(1):1–43.
- McNeil, A. J. and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300.
- Muler, N. and Yohai, V. J. (2008). Robust estimates for GARCH models. *Journal of Statistical Planning and Inference*, 138(10):2918–2940.
- Nareklashvili, M., Polson, N. G., and Sokolov, V. (2025). Generative quantile Bayesian prediction. *arXiv preprint arXiv:2510.21784*.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.
- Nieto, M. R. and Ruiz, E. (2016). Frontiers in VaR forecasting and backtesting. *International Journal of Forecasting*, 32(2):475–501.

- Nolde, N. and Ziegel, J. F. (2017). Elicitability and backtesting: Perspectives for banking regulation. *Annals of Applied Statistics*, 11(4):1833–1874.
- Patton, A. J. (2020). Comparing possibly misspecified forecasts. *Journal of Business and Economic Statistics*, 38(4):796–809.
- Patton, A. J., Ziegel, J. F., and Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and Value-at-Risk). *Journal of Econometrics*, 211(2):388–413.
- Pérignon, C. and Smith, D. R. (2010). The level and quality of value-at-risk disclosure by commercial banks. *Journal of Banking and Finance*, 34(2):362–377.
- Pickands, J. (1975). Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1):119–131.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.
- Polson, N. G. and Sokolov, V. (2023). Generative AI for Bayesian computation. *arXiv preprint arXiv:2305.14972*.
- Qu, R., Timmermann, A., and Zhu, Y. (2024). Comparing forecasting performance with panel data. *International Journal of Forecasting*, 40(3):918–941.
- Romano, J. P. and Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282.
- Romano, J. P. and Wolf, M. (2016). Efficient computation of adjusted p -values for resampling-based stepdown multiple testing. *Statistics and Probability Letters*, 113:38–40.
- Romano, Y., Patterson, E., and Candès, E. J. (2019). Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 3538–3548.
- Shumway, T. (1997). The delisting bias in CRSP data. *Journal of Finance*, 52(1):327–340.
- Steinwart, I. and Christmann, A. (2011). Estimating conditional quantiles with the help of the pinball loss. *Bernoulli*, 17(1):211–225.
- Taylor, J. W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric Laplace distribution. *Journal of Business and Economic Statistics*, 37(1):121–133.
- Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Zhang, C., Zhang, Y., Cucuringu, M., and Qian, Z. (2024). Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22(2):492–530.